

# ST MC SDK 5.x 相电流检测与重构 位置、速度信息获取

STM32电动机控制应用系列讲座之三



## □ 相电流的检测与重构

- ICS
- 三电阻
- 单电阻

## □ 转子位置、速度信息的获取

- 有位置传感器
- 无位置速度传感器

# 相电流的检测与重构



## □ 相电流的检测与重构

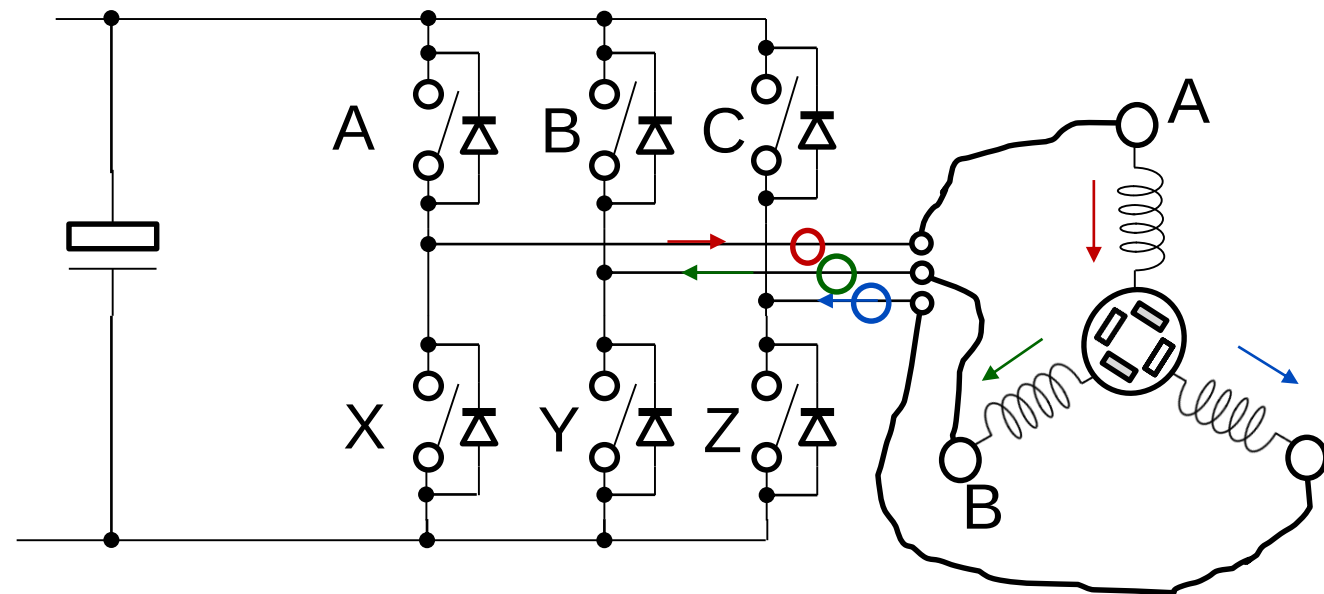
- ICS
- 三电阻
- 单电阻

## □ 转子位置、速度信息的获取

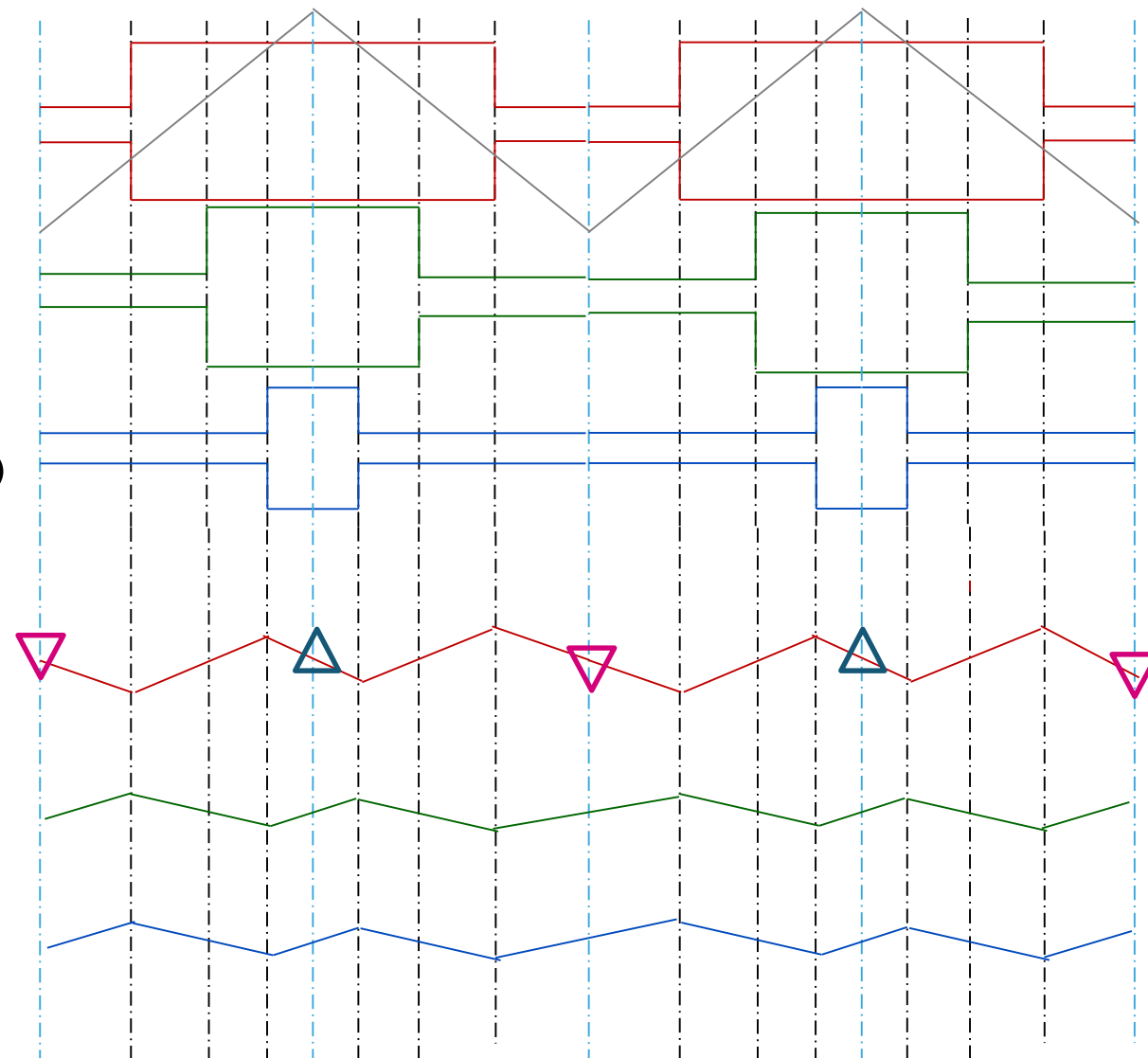
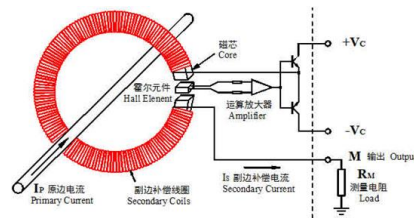
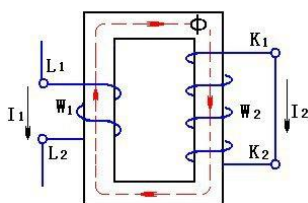
- 有位置传感器
- 无位置速度传感器

# 电流采样 — ICS

4



|      | ACCT             | DCCT        |
|------|------------------|-------------|
| 频率范围 | >0 Hz ~ tens kHz | DC ~ 100kHz |
| 退磁   | 需要               | 不需要         |
| 成本   | 低                | 高           |



# ST MC SDK5.x ICS采样固件

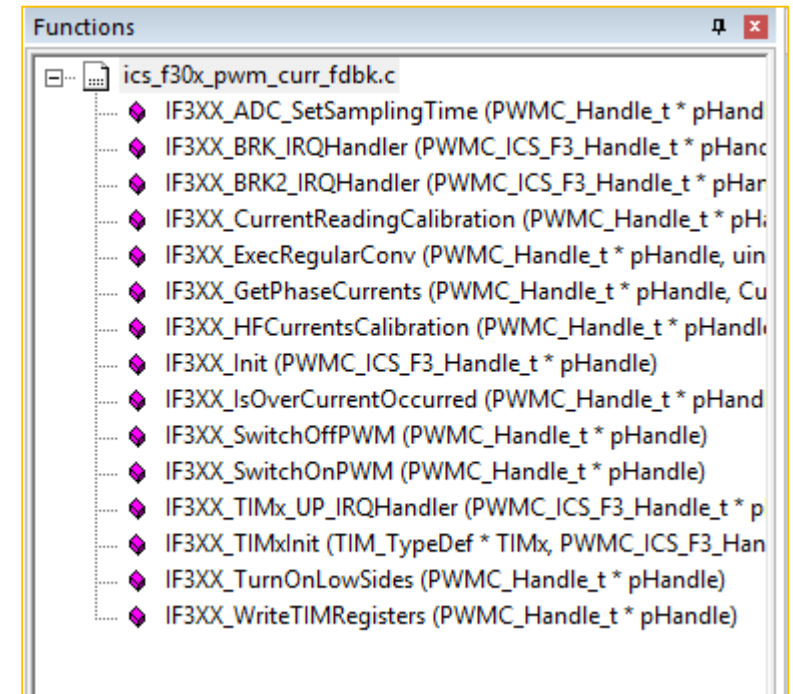
5

## ➤ 针对STM32系列芯片都有各自的文件:

- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F1xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F3xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F4xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F7xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\L4xx\Src

## ➤ 文件名称

- ✓ ics\_yxx\_pwm\_curr\_fdbk.c
- ✓ y=f1, f3, f4, f7, l4



# 相电流的检测与重构



## □ 相电流的检测与重构

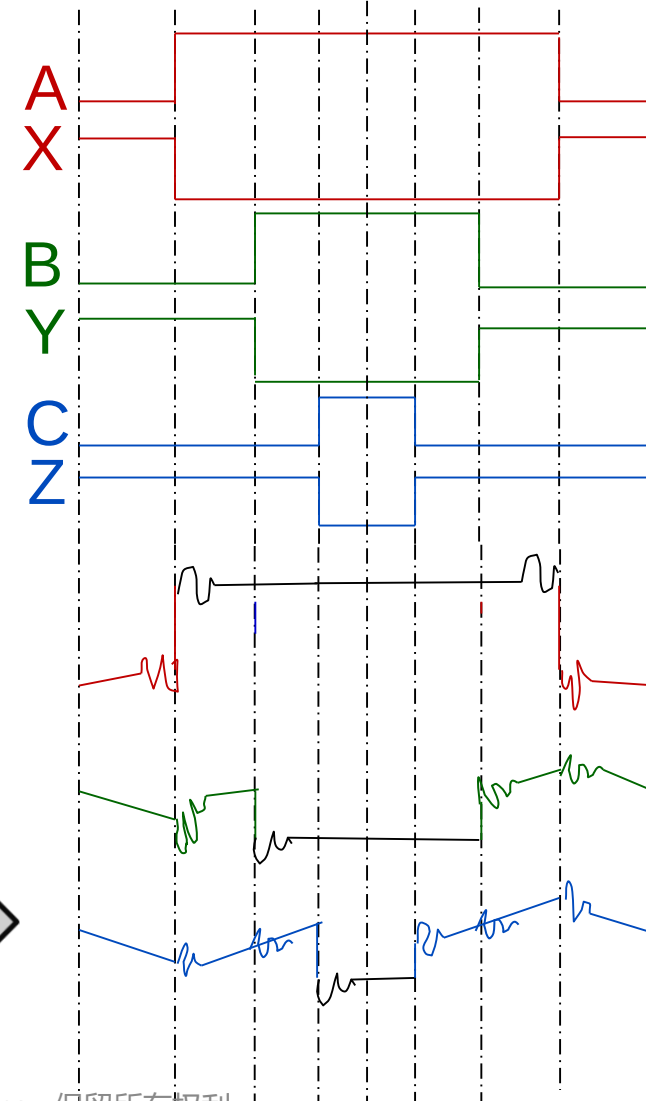
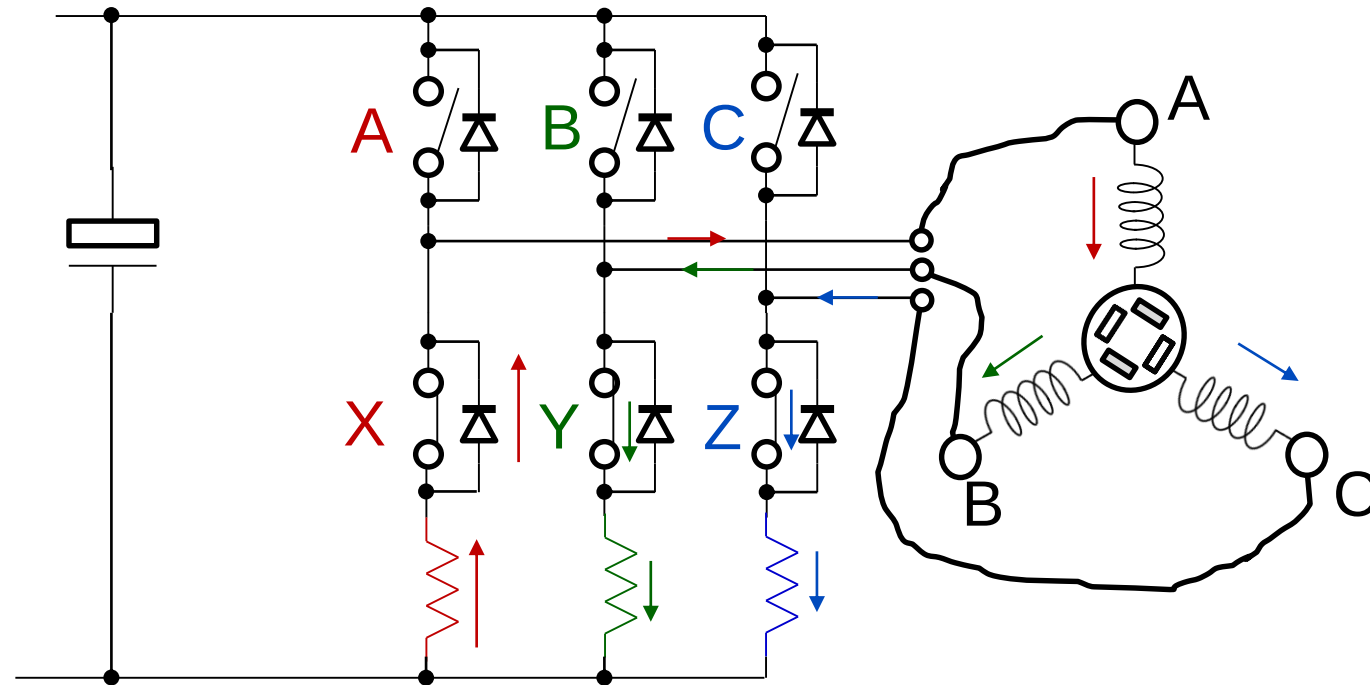
- ICS
- 三电阻
- 单电阻

## □ 转子位置、速度信息的获取

- 有位置传感器
- 无位置速度传感器

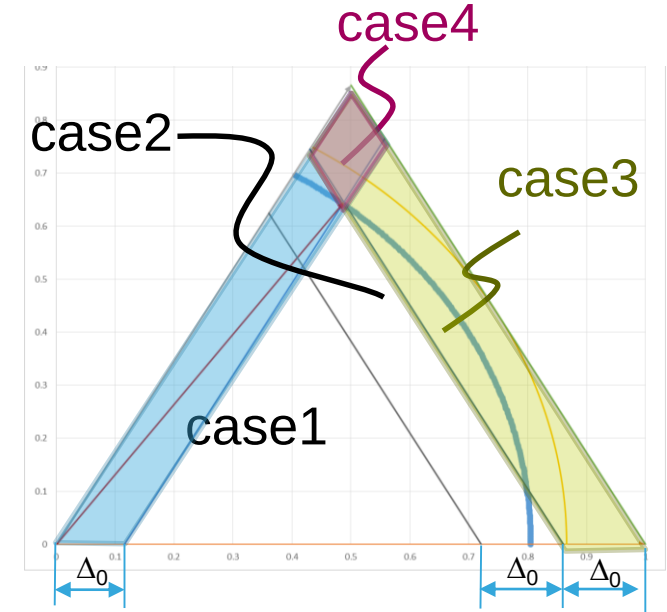
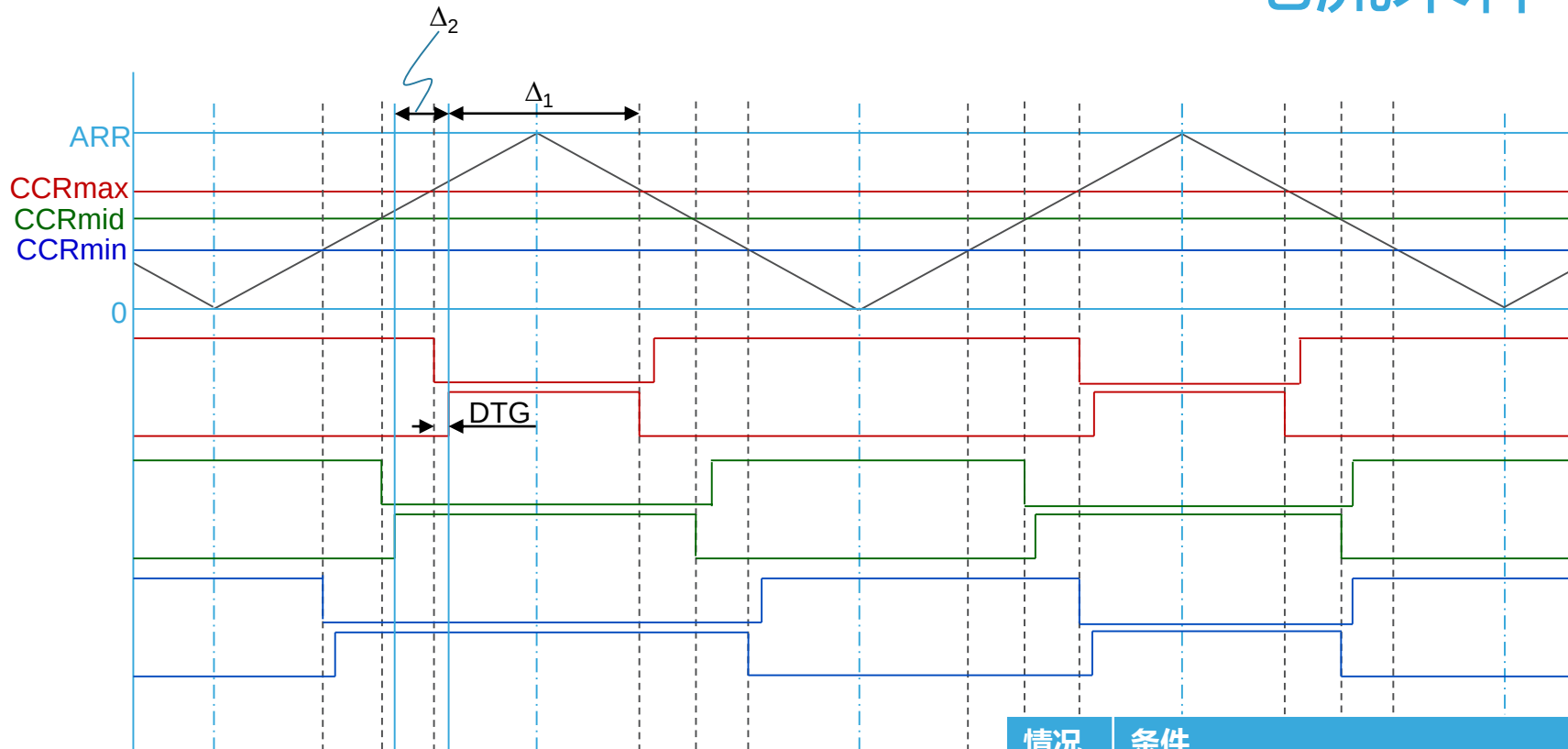
# 电流采样 — 三电阻(1/2)

7



# 电流采样 — 三电阻(2/2)

8



$$\Delta_1 = 2 * (ARR - CCRmax - DTG)$$

$$\Delta_2 = CCRmax - CCRmid$$

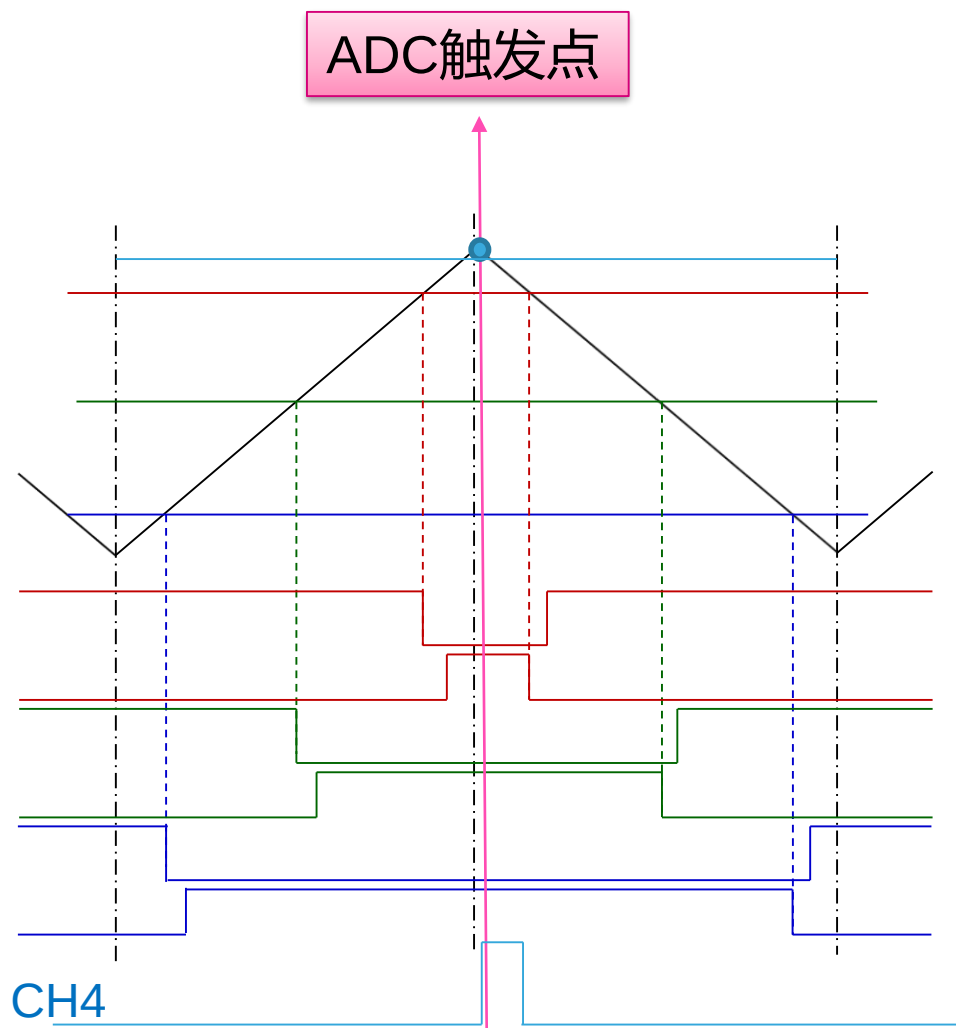
$$\Delta_0 = CNT\_Ton + CNT\_Tring + CNT\_TADCSH(COV), \\ Tring > TADCsta$$

$$\Delta_0 = CNT\_Ton + CNT\_TADCsta + CNT\_TADCSH(COV), TADCsta \geq Tring$$

| 情况 | 条件                                                                                                              | 采样点                                       |
|----|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 1  | $\Delta_1 > \max(2 * (CNT\_Ton + CNT\_Trise + CNT\_Ring + tdead/2), CNT\_TADCsta + CNT\_TADCSH(COV) - tdead/2)$ | Middle of PWM                             |
| 2  | $\Delta_1 > \Delta_0$                                                                                           | $CCRmax + tdead + ton + tring + \epsilon$ |
| 3  | $\Delta_2 > \Delta_0 > \Delta_1$                                                                                | $CCRmid + tdead + ton + tring + \epsilon$ |
| 4  | $\Delta_1 < \Delta_0$ and $\Delta_2 < \Delta_0$                                                                 | Not available                             |

# 三电阻ADC触发机制说明

9



- 配置TIMER CH4为TRGO输出
- TIMER的TRGO输出用于硬件触发ADC采样;
- 如果是拥有两个ADC模块的可以同时触发进行同时采样
- 判断波形的采样位置, 修正CCR4数据

# ST MC SDK5.x 三电阻采样固件

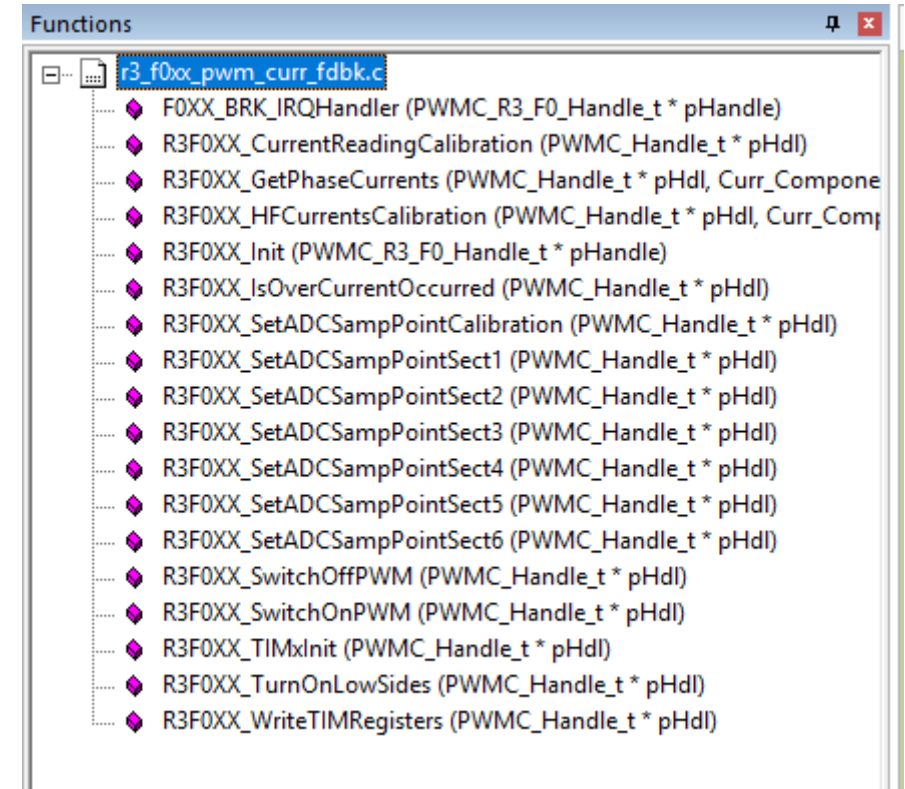
10

## ❑ 针对STM32系列芯片都有各自的文件：

1. xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F0xx\Src
2. xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F1xx\Src
3. xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F3xx\Src
4. xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F4xx\Src
5. xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F7xx\Src
6. xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\L4xx\Src

## ❑ 文件名称

- r3\_z\_yxx\_pwm\_curr\_fdbk.c
  - ✓ y=f0, f1, f3, f4, f7, l4
  - ✓ z=1,4(for dual motor),[].s



# 相电流的检测与重构

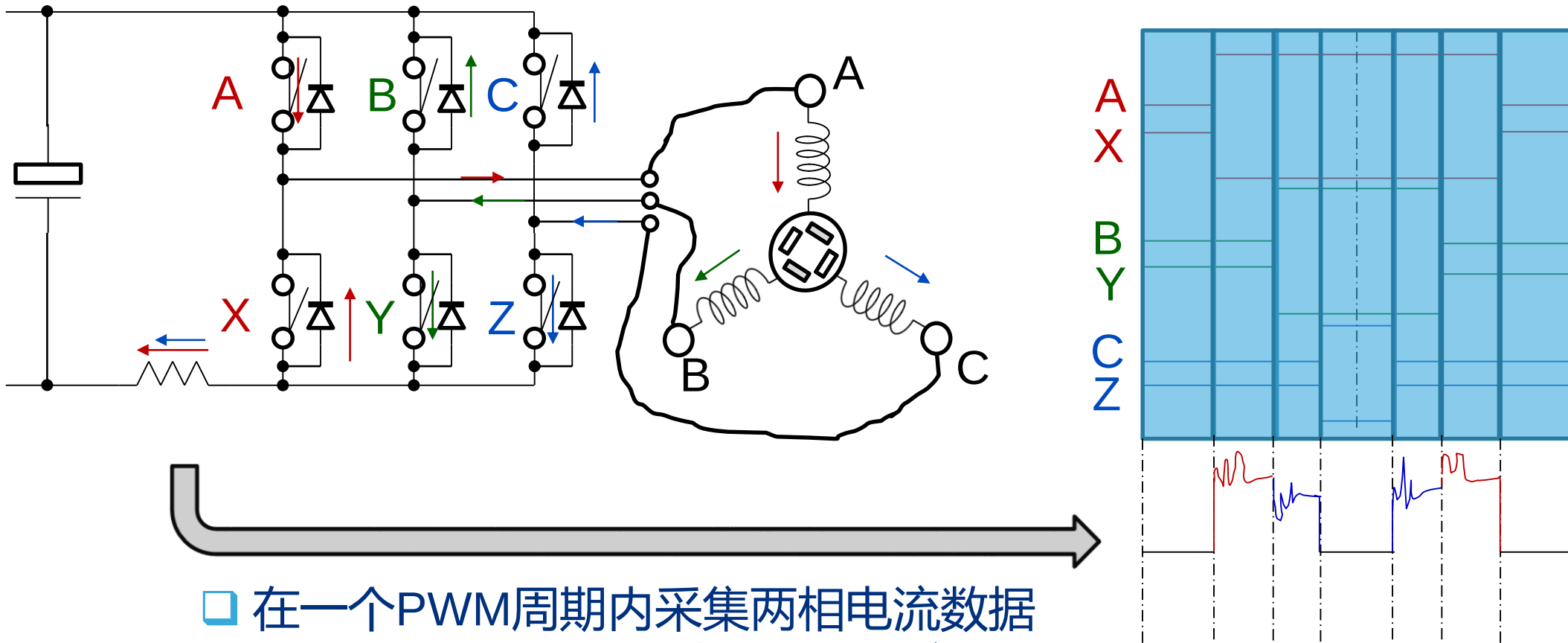


## □ 相电流的检测与重构

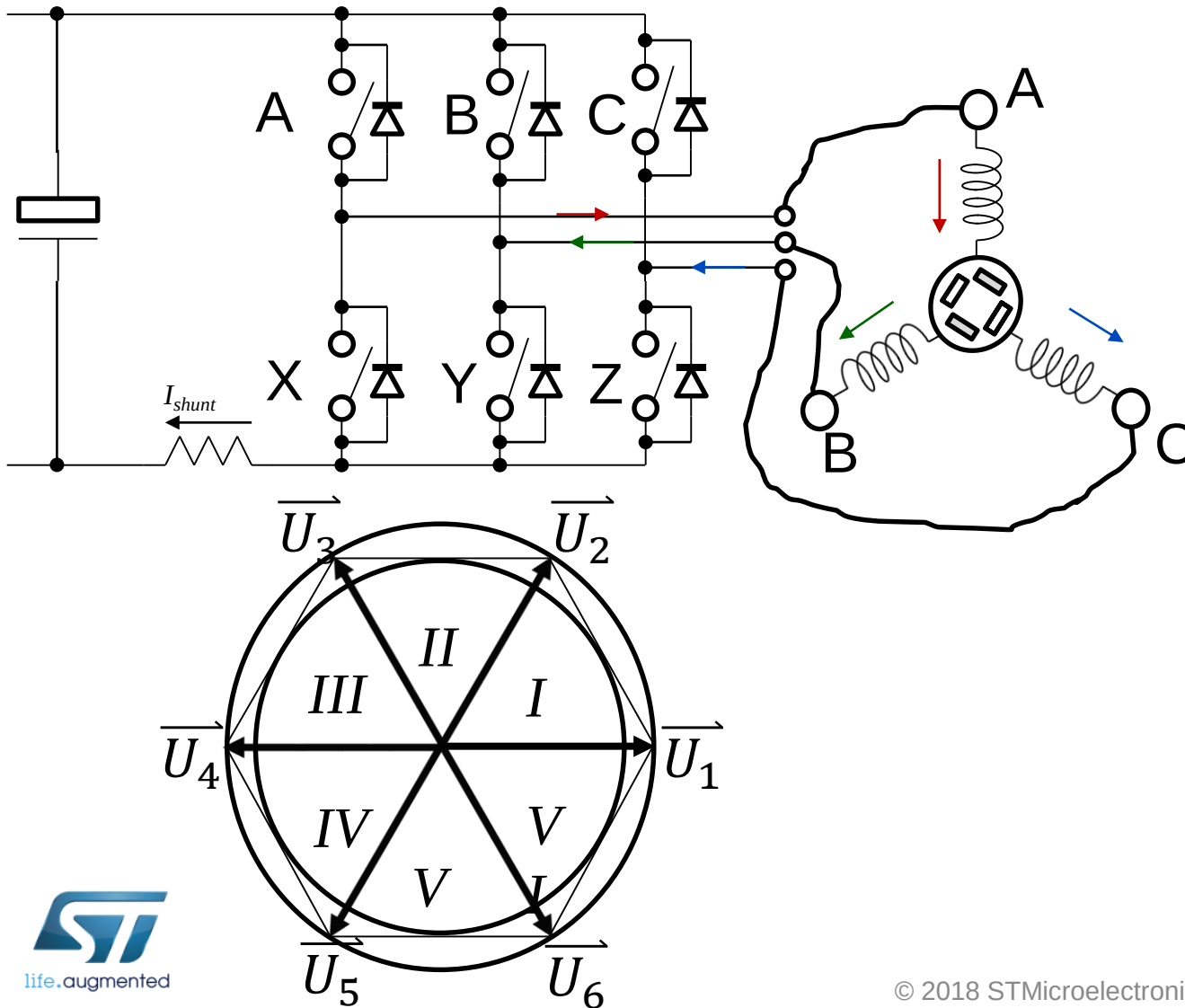
- ICS
- 三电阻
- 单电阻

## □ 转子位置、速度信息的获取

- 有位置传感器
- 无位置速度传感器

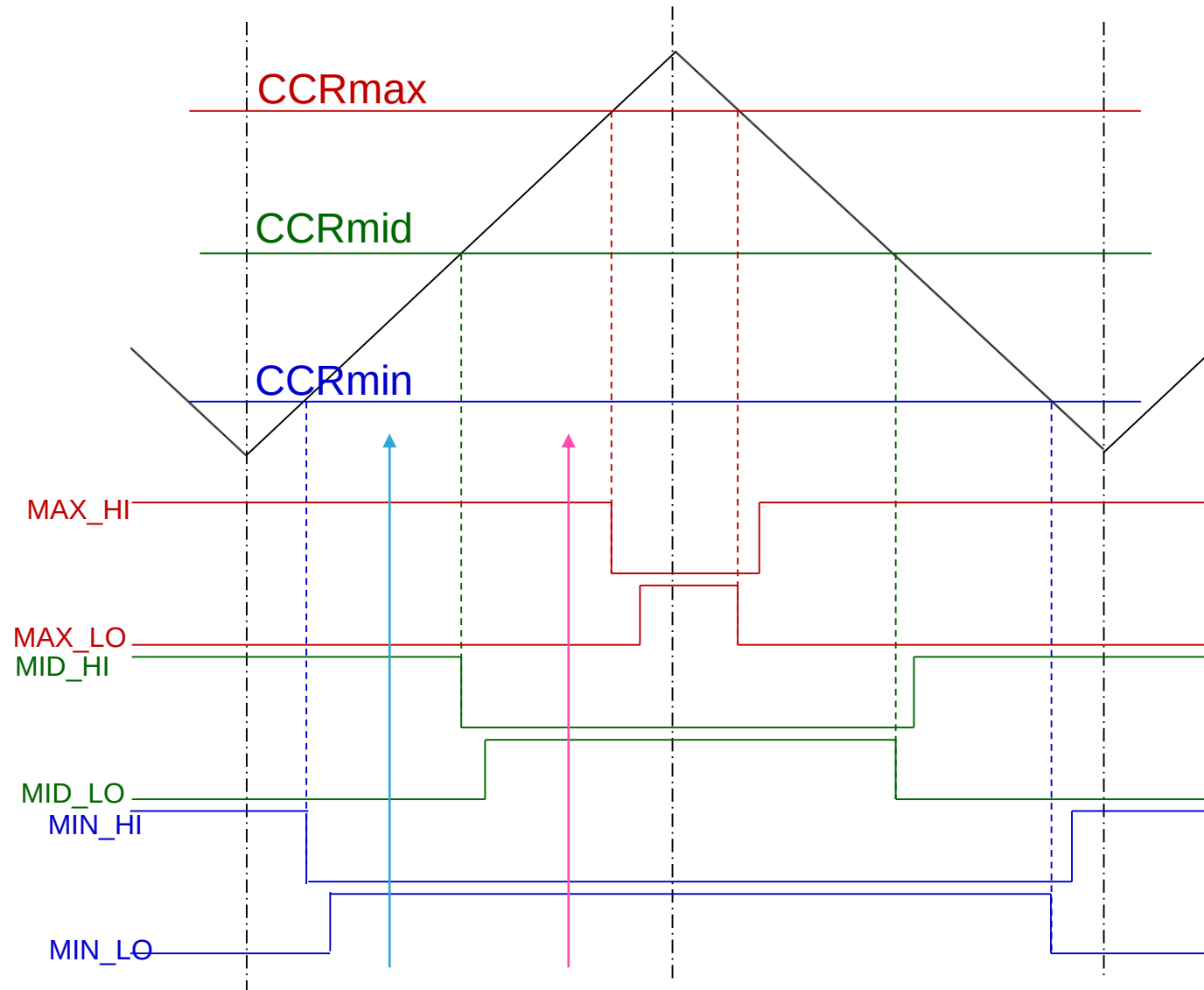


- 在一个PWM周期内采集两相电流数据
- 可根据 $I_a + I_b + I_c = 0$ 构造出三相电流



□ 空间电压矢量与单电阻检测到的信号对应的相电流的关系

| Vector      | A(X)    | B(Y)    | C(Z)    | $I_{shunt}$ |
|-------------|---------|---------|---------|-------------|
| $\vec{U}_0$ | OFF(ON) | OFF(ON) | OFF(ON) | 0           |
| $\vec{U}_1$ | ON(OFF) | OFF(ON) | OFF(ON) | $i_A$       |
| $\vec{U}_2$ | ON(OFF) | ON(OFF) | OFF(ON) | $-i_C$      |
| $\vec{U}_3$ | OFF(ON) | ON(OFF) | OFF(ON) | $i_B$       |
| $\vec{U}_4$ | OFF(ON) | ON(OFF) | ON(OFF) | $-i_A$      |
| $\vec{U}_5$ | OFF(ON) | OFF(ON) | ON(OFF) | $i_C$       |
| $\vec{U}_6$ | ON(OFF) | OFF(ON) | ON(OFF) | $-i_B$      |
| $\vec{U}_7$ | ON(OFF) | ON(OFF) | ON(OFF) | 0           |



$$T_p = T_{dead} + T_{on} + T_{ADC\ s/h} + \text{Max}(T_{ring}, T_{ADCtrigger\ delay})$$

|                         |              |
|-------------------------|--------------|
| $T_p$                   | – 电流采样最小脉宽时间 |
| $T_{dead}$              | – 死区时间       |
| $T_{on}$                | – ADC启动时间    |
| $T_{ring}$              | – 振铃时间       |
| $T_{ADCtrigger\ delay}$ | – ADC触发延迟时间  |
| $T_{ADCs/h}$            | – ADC采样时间    |

ADC采样触发点:

第一点:

当  $T(\text{CCRmid} - \text{CCRmin}) > T_{dead} + T_{ring} + T_{ADC\ s/h}$

采样点为  $\frac{T(\text{CCRmid} + \text{CCRmin}) + T_{dead}}{2}$

否则为  $T_{CCRmid} - T_{ADCtrigger\ delay} - T_{ADC\ s/h}$

第二点:

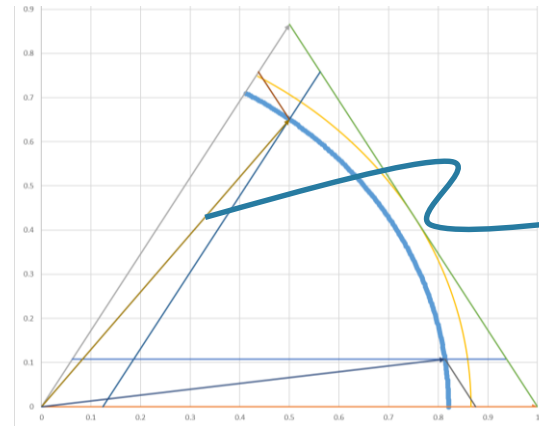
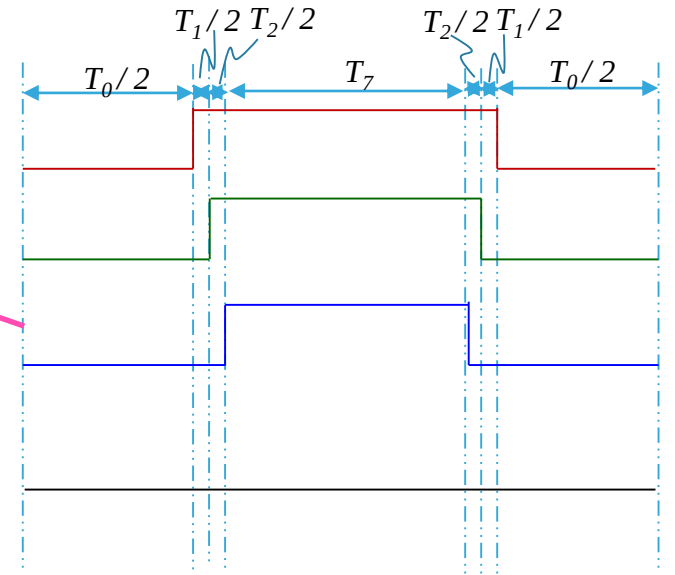
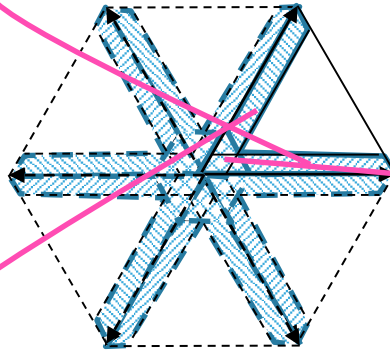
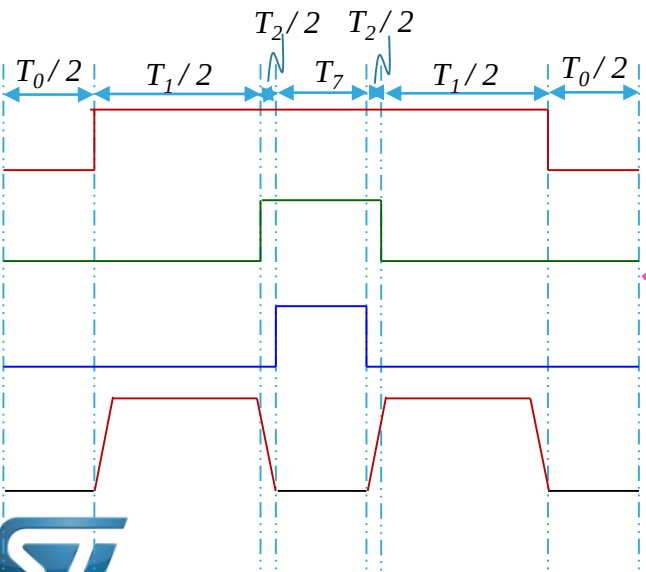
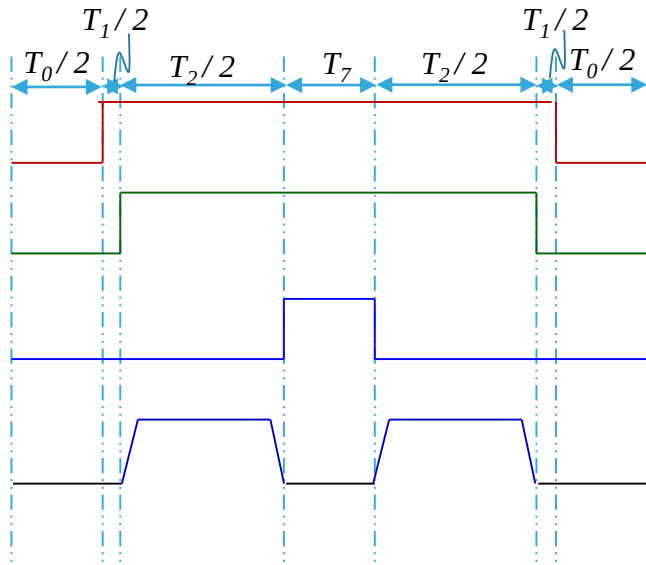
当  $T(\text{CCRmax} - \text{CCRmid}) > T_{dead} + T_{ring} + T_{ADC\ s/h}$

采样点为  $\frac{T(\text{CCRmax} + \text{CCRmid}) + T_{dead}}{2}$

否则为  $T_{CCRmax} - T_{ADCtrigger\ delay} - T_{ADC\ s/h}$

# 电流采样 — 单电阻(4/8)

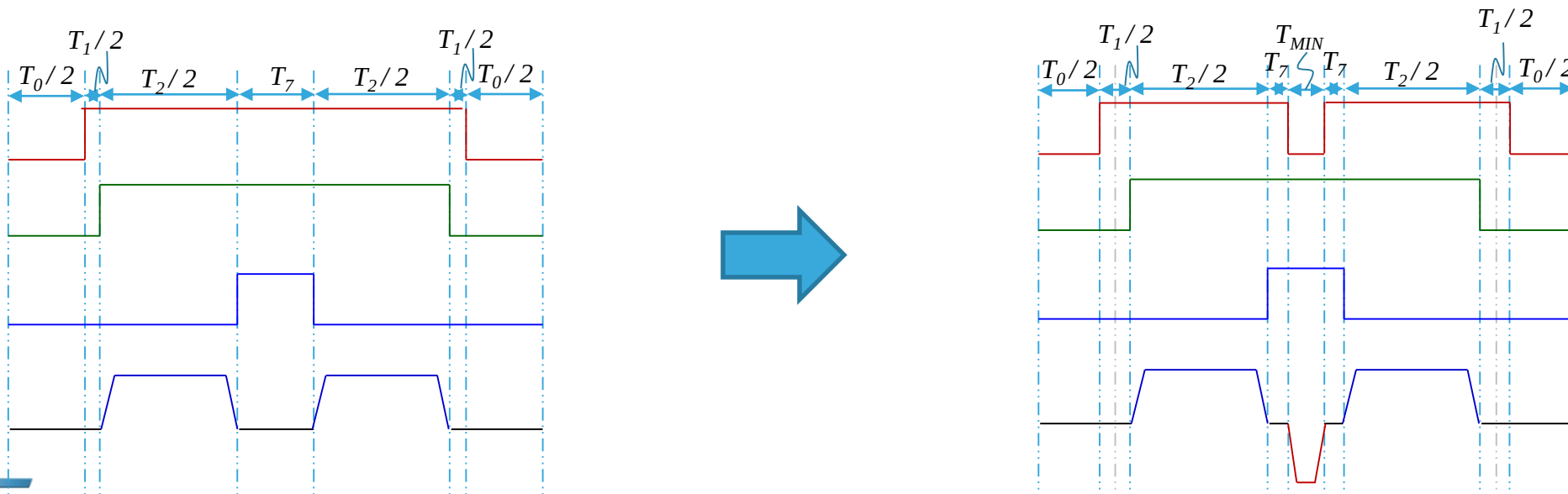
15



MMI=Max Modulation Index

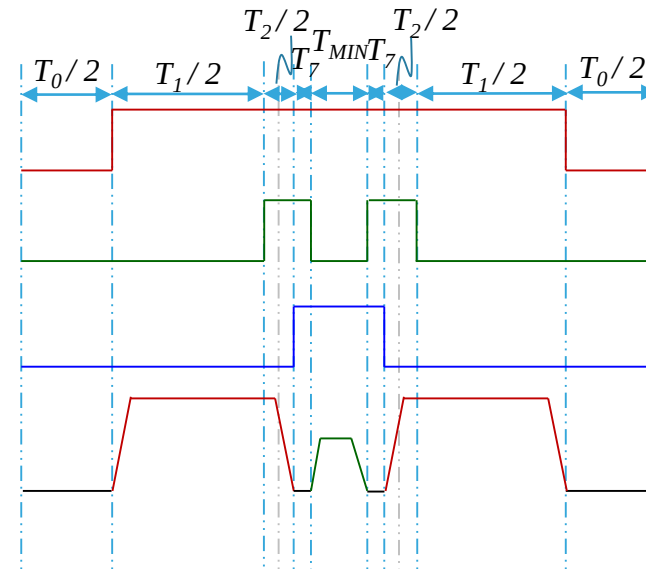
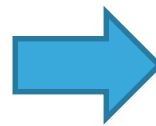
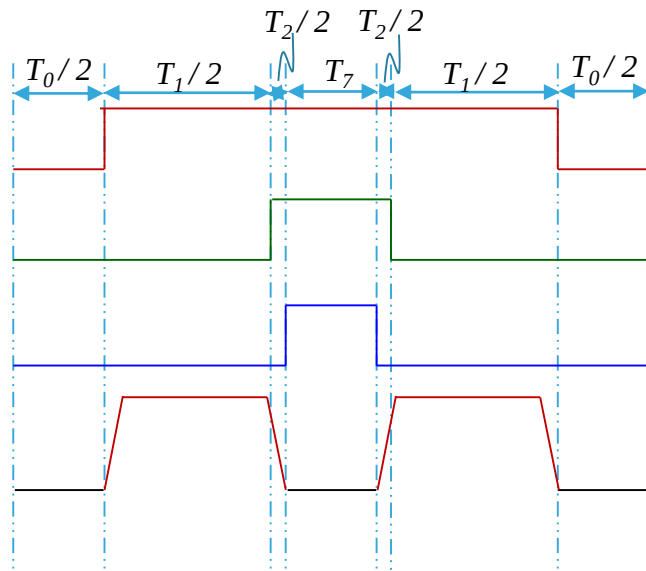
ST 专利  
US20090284194 A1

ST 专利 (Pat. Pub. No.: US20090284194 A1) 解决单电阻无法采样问题



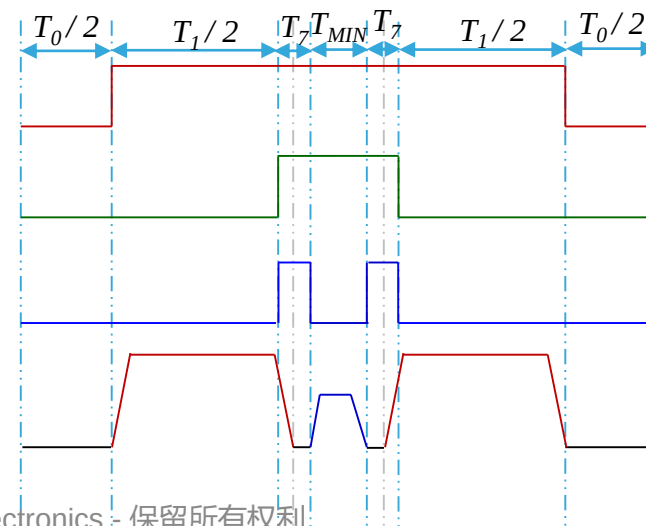
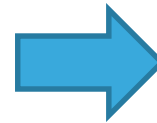
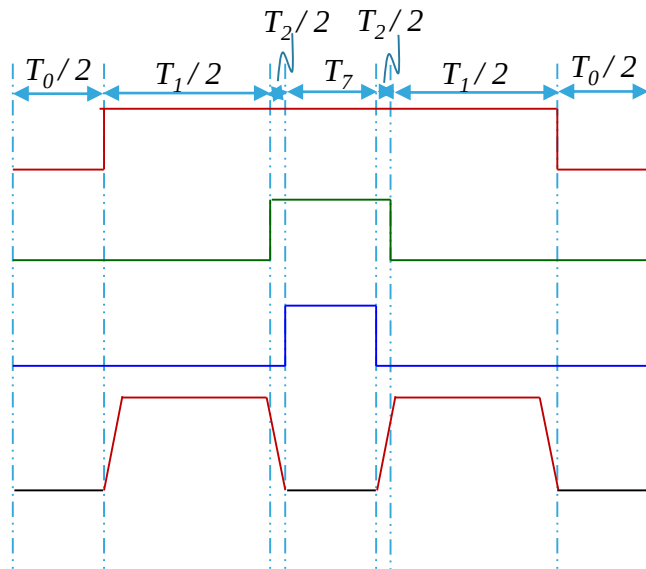
ST 专利  
US20090284194 A1

ST 专利 (Pat. Pub. No.: US20090284194 A1) 解决单电阻无法采样问题



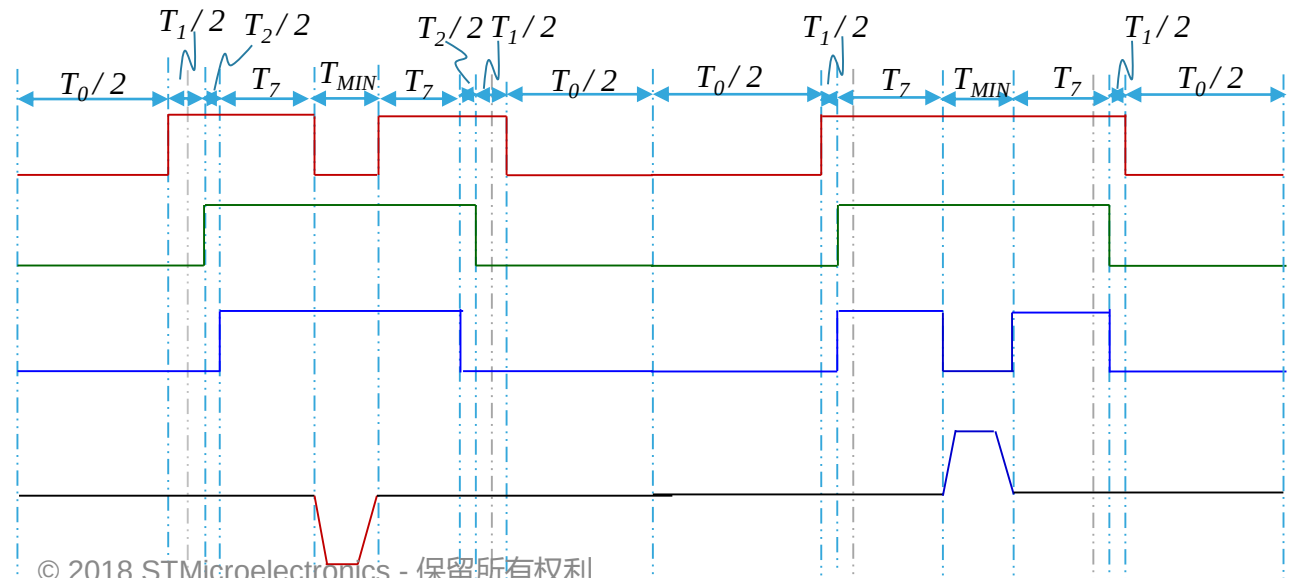
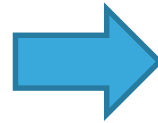
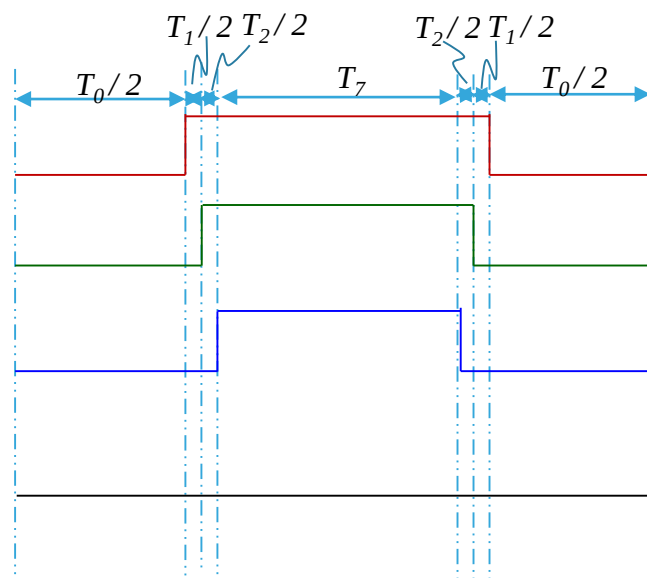
ST 专利  
US20090284194 A1

ST 专利 (Pat. Pub. No.: US20090284194 A1) 解决单电阻无法采样问题



ST 专利  
US20090284194 A1

ST 专利 (Pat. Pub. No.: US20090284194 A1) 解决单电阻无法采样问题



# ADC采样机制说明

20

|                     |                                                                                            |             |     |   |       |             |
|---------------------|--------------------------------------------------------------------------------------------|-------------|-----|---|-------|-------------|
| $t_{latr}^{(1)}$    | Trigger conversion latency<br>Regular and injected<br>channels without conversion<br>abort | CKMODE = 00 | 1.5 | 2 | 2.5   | $1/f_{ADC}$ |
|                     |                                                                                            | CKMODE = 01 | -   | - | 2     | $1/f_{ADC}$ |
|                     |                                                                                            | CKMODE = 10 | -   | - | 2.25  | $1/f_{ADC}$ |
|                     |                                                                                            | CKMODE = 11 | -   | - | 2.125 | $1/f_{ADC}$ |
| $t_{latrinj}^{(1)}$ | Trigger conversion latency<br>Injected channels aborting a<br>regular conversion           | CKMODE = 00 | 2.5 | 3 | 3.5   | $1/f_{ADC}$ |
|                     |                                                                                            | CKMODE = 01 | -   | - | 3     | $1/f_{ADC}$ |
|                     |                                                                                            | CKMODE = 10 | -   | - | 3.25  | $1/f_{ADC}$ |
|                     |                                                                                            | CKMODE = 11 | -   | - | 3.125 | $1/f_{ADC}$ |

可在对应芯片数  
据手册中查询到  
对应参数

触发转换延迟时间

ADC采样保持时间

ADC转换时间

触发沿可选择,  
触发源可选

可配置采保时间

ADC转换时间根  
据ADC精度而定

上升沿

下降沿

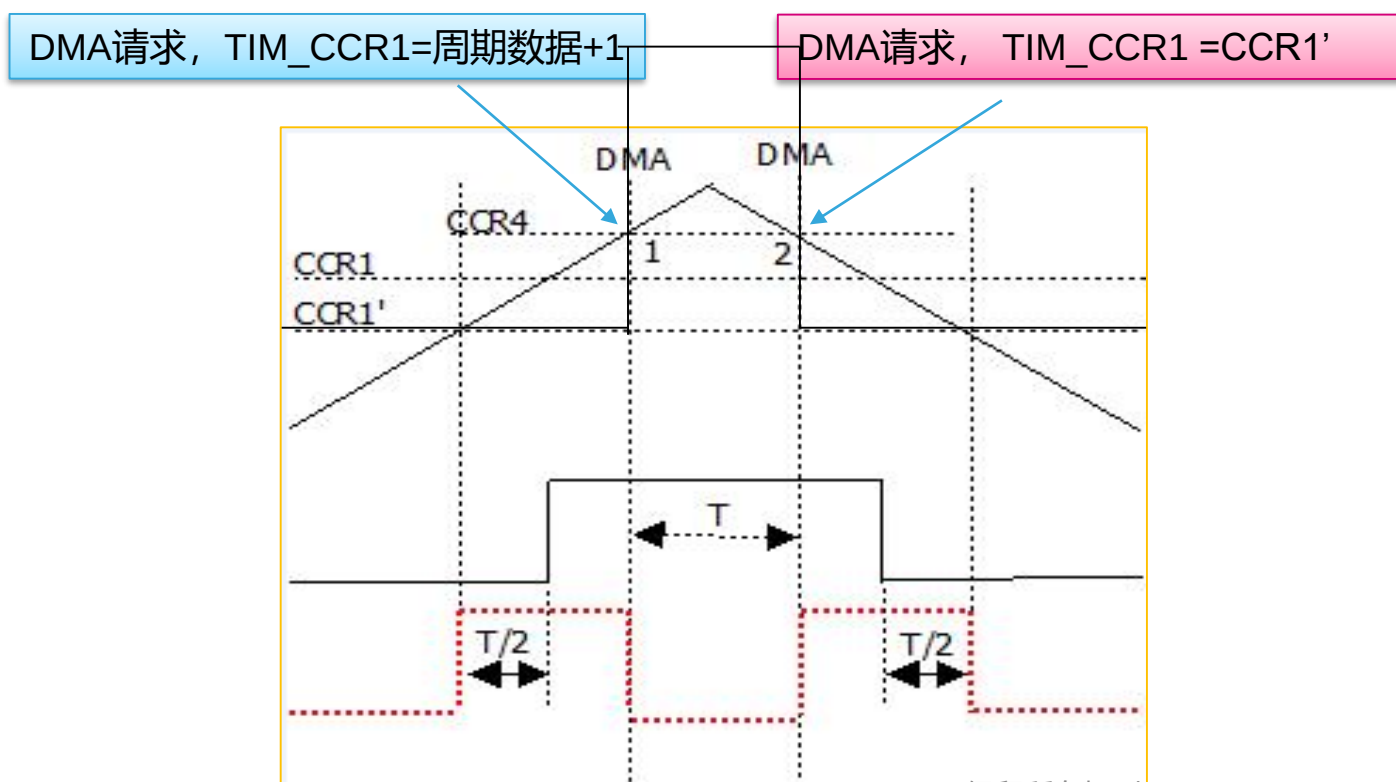
上升/下降沿

| Name   | Source           |
|--------|------------------|
| JEXT0  | TIM1_TRGO event  |
| JEXT1  | TIM1_CC4 event   |
| JEXT2  | TIM2_TRGO event  |
| JEXT3  | TIM2_CC1 event   |
| JEXT4  | TIM3_CC4 event   |
| JEXT5  | TIM4_TRGO event  |
| JEXT6  | EXTI line 15     |
| JEXT7  | Reserved         |
| JEXT8  | TIM1_TRGO2 event |
| JEXT9  | Reserved         |
| JEXT10 | Reserved         |
| JEXT11 | TIM3_CC3 event   |
| JEXT12 | TIM3_TRGO event  |

# 波形变形机制说明

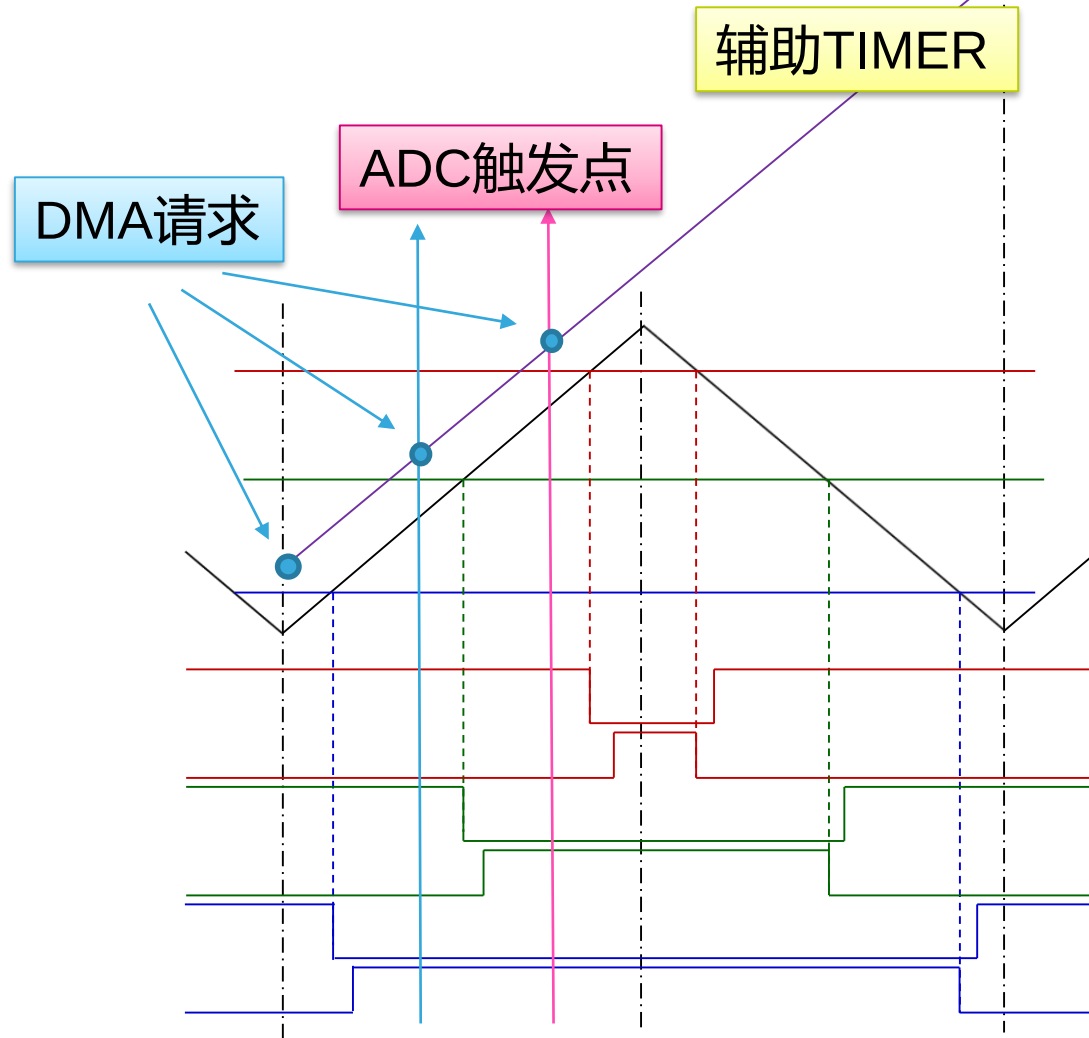
21

- $CCR4 = T_c/2 - \min Pulse/2$  在CCR4 比较值1部分产生DMA事件; Timer1的preload为禁止状态;  $CCR1 = T_c/2 + \delta$
- 在CCR4 比较值2 部分产生DMA事件;  $CCR1' = CCR1 - \min Pulse/2$



# 单电阻ADC触发机制说明

22



- 辅助TIMER（比如TIM15）与电机TIMER（如TIM1）做同步动作，同频输出；
- 配置DMA重载辅助TIMER的CCR寄存器；
- 辅助TIMER的TRGO输出用于硬件触发ADC采样；
- ADC配置为非连续模式，序列采样中断，采样两次后进入FOC处理中断中；

*在STM32F3xx系列中电机TIMER有自带的机制可以灵活配置*

# ST MC SDK5.x 单电阻采样固件

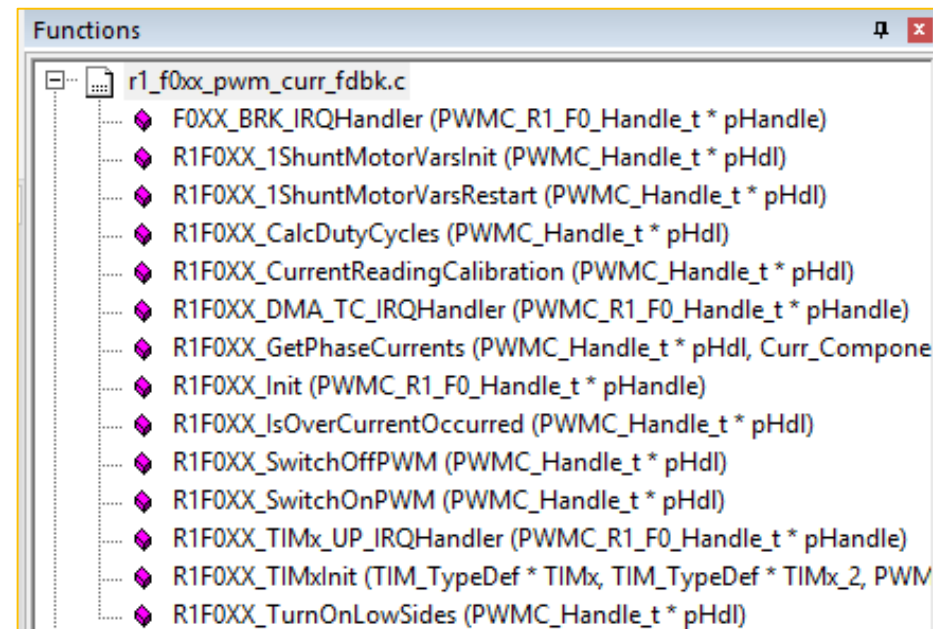
23

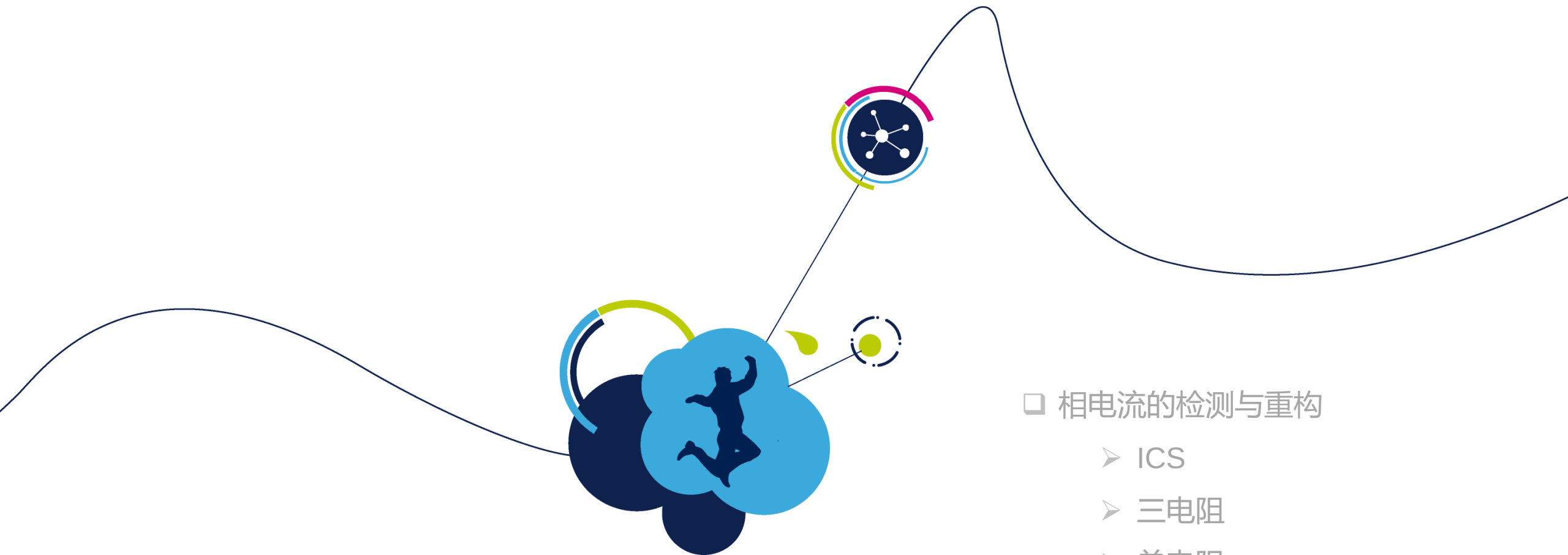
## ➤ 针对STM32系列芯片都有各自的文件:

- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F0xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F1xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F3xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F4xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\F7xx\Src
- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\L4xx\Src

## ➤ 文件名称

- ✓ r1\_yxx\_pwm\_curr\_fdbk.c
- ✓ y=f0, f1, f3, f4, f7, l4





# 转子位置、速度信息的获取

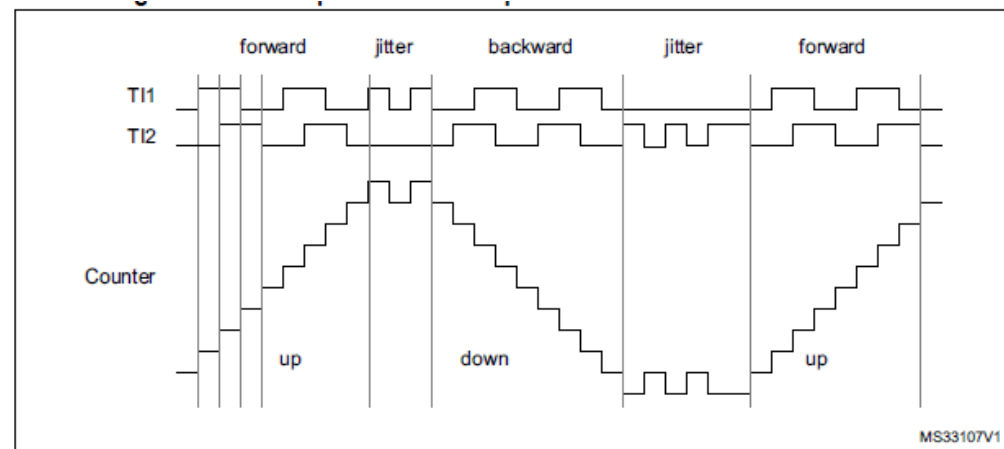
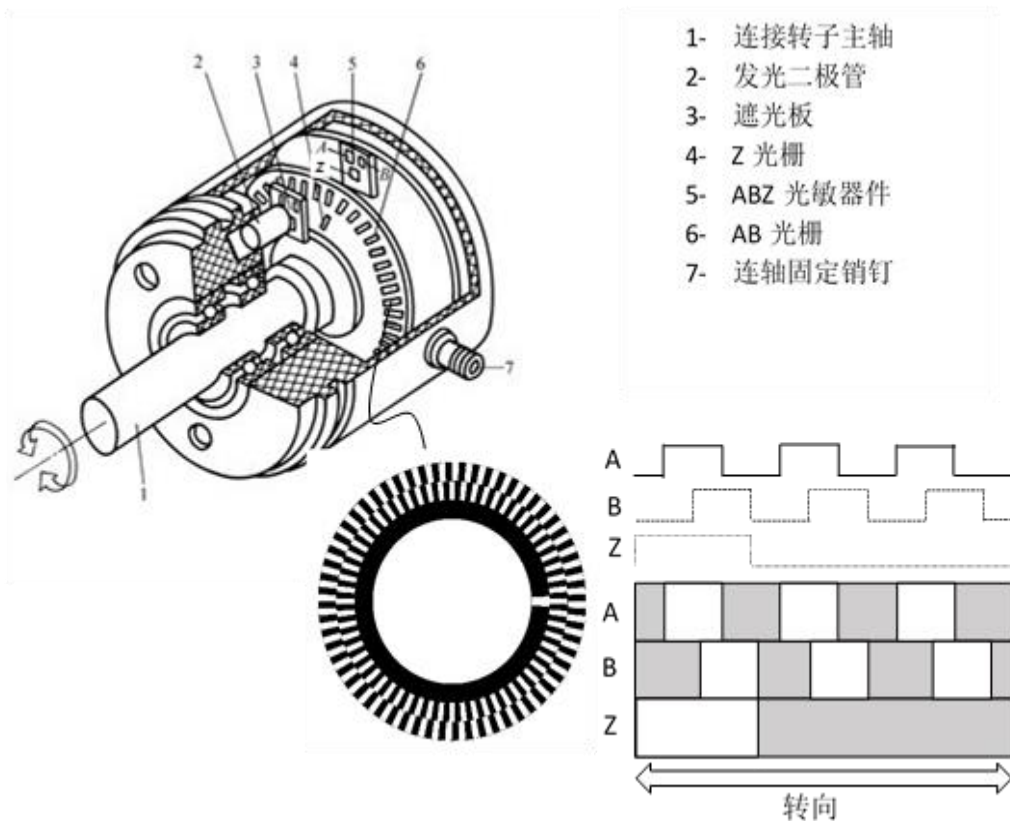
## □ 相电流的检测与重构

- ICS
- 三电阻
- 单电阻

## □ 转子位置、速度信息的获取

- 有位置传感器
- 无位置速度传感器

# 位置速度检测 — Encoder 25

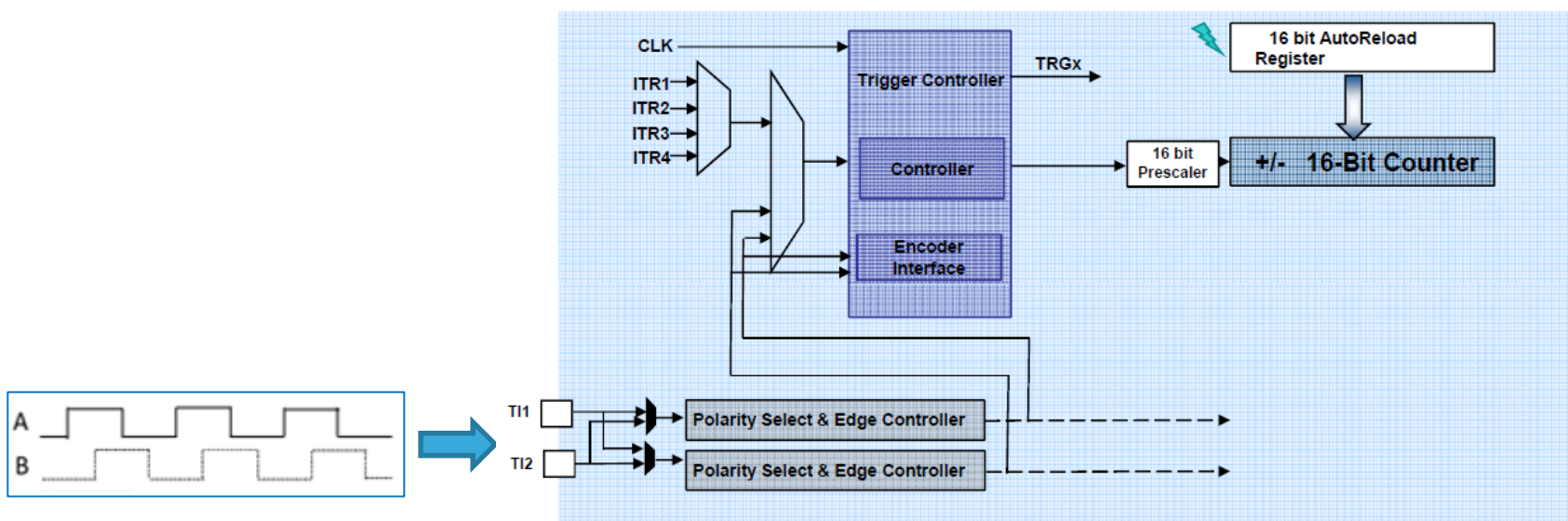


- 使用增量编码器时，在第一次电机启动，任意保护停止或者MCU复位后都要进行预定位操作。
- Z信号（一圈一个）可以使用外部中断或者外部Timer捕捉模式，代表编码器的0度位置，可以用于校准角度位置，可以使用DMA模式对编码器模块赋值；

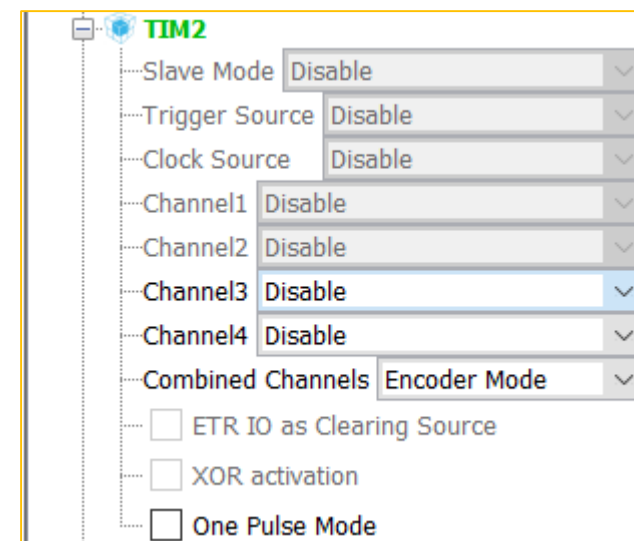
# STM32硬件Encoder接口

26

- 在全系列STM32中都有硬件增量编码器Encoder接口
- 每个正交沿都可为加/减计数



CubeMx中的配置



# ST MC SDK5.x Encoder固件

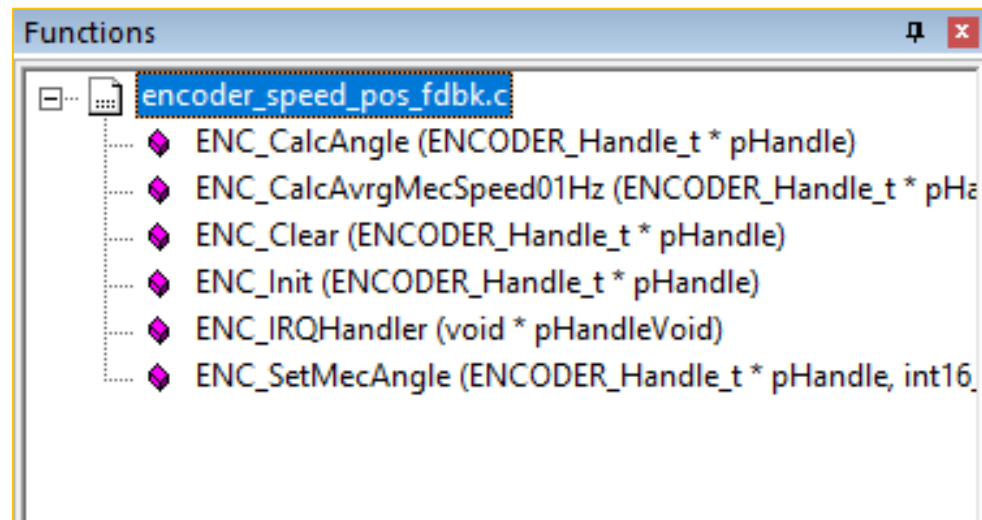
27

## ➤ 具体文件夹如下:

- ✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\Any\Src

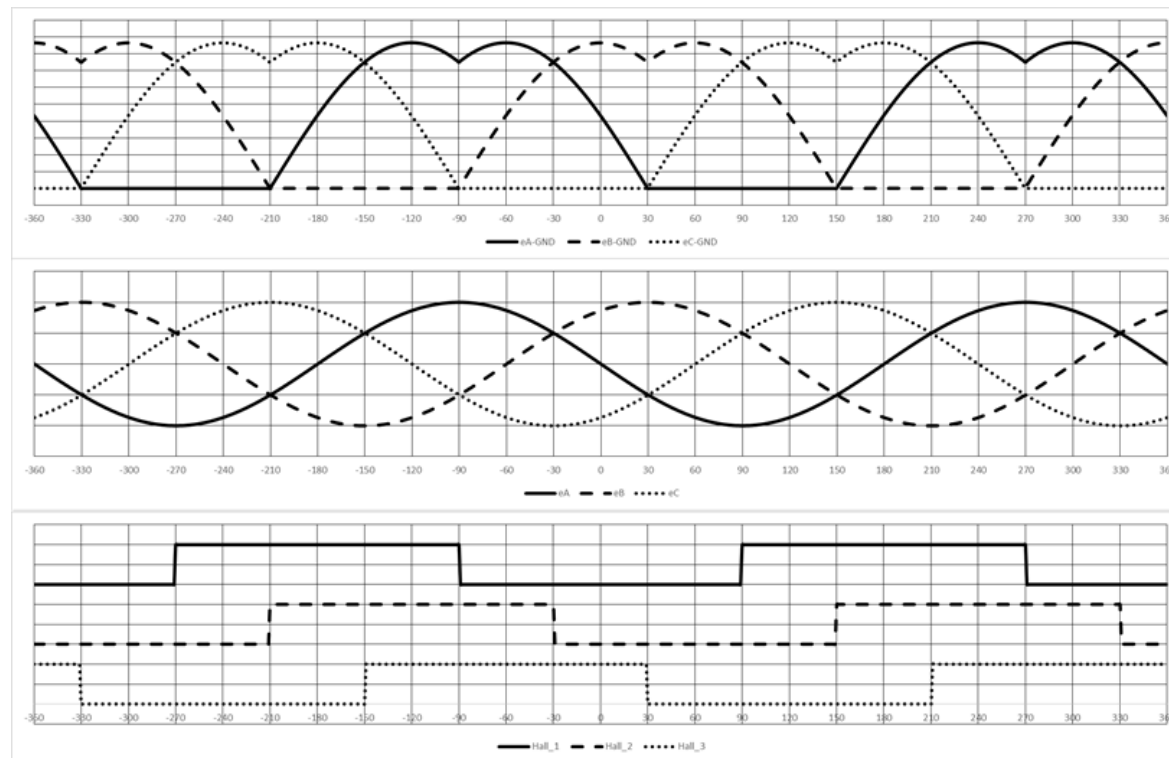
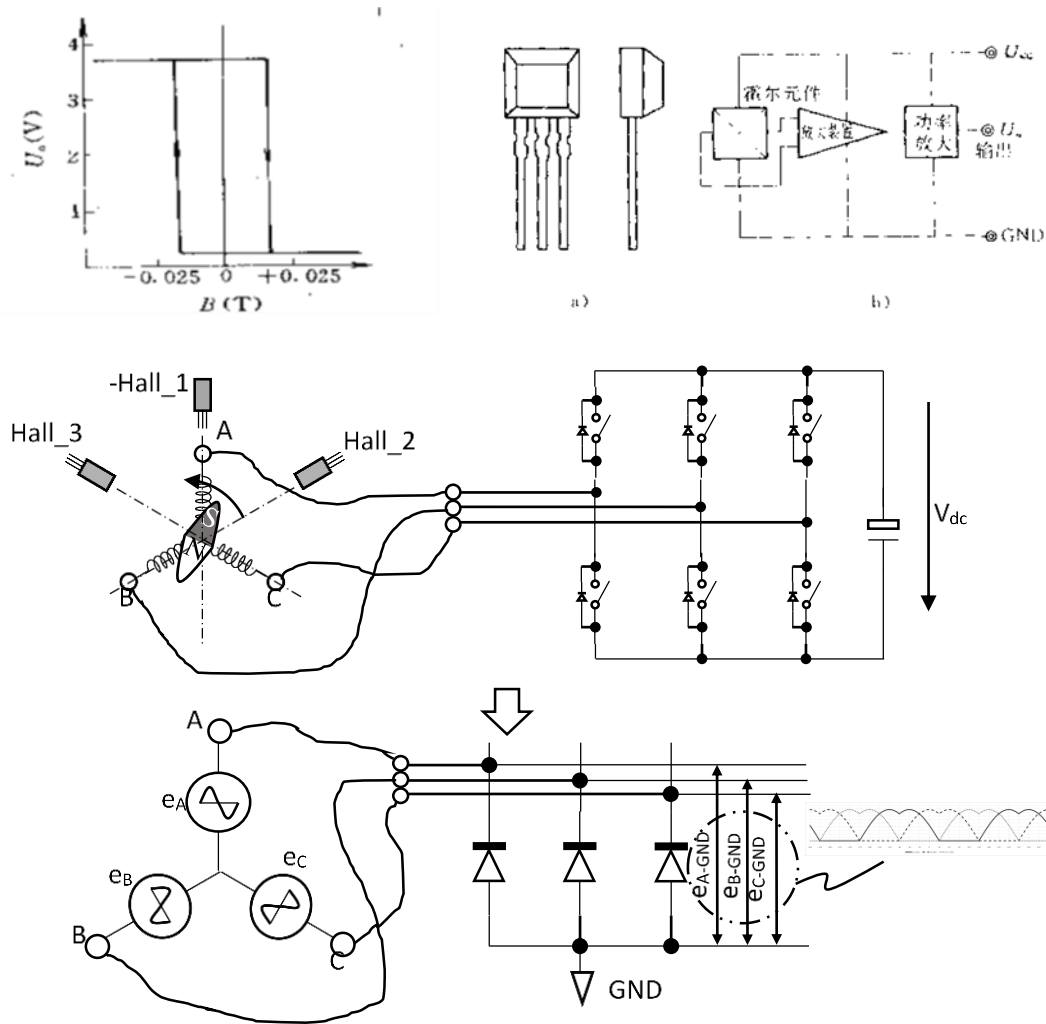
## ➤ 文件名称

- ✓ encoder\_speed\_pos\_fdbk.c



# 位置速度检测 — Hall传感器

28

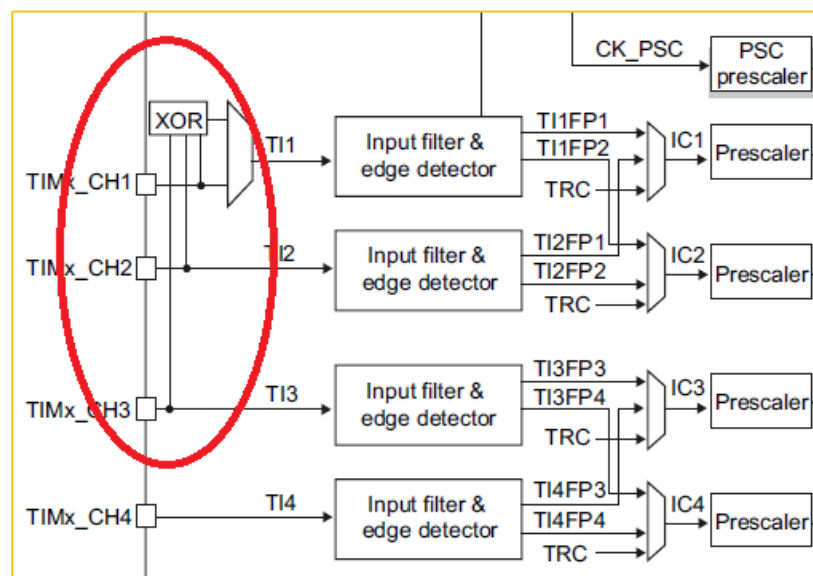
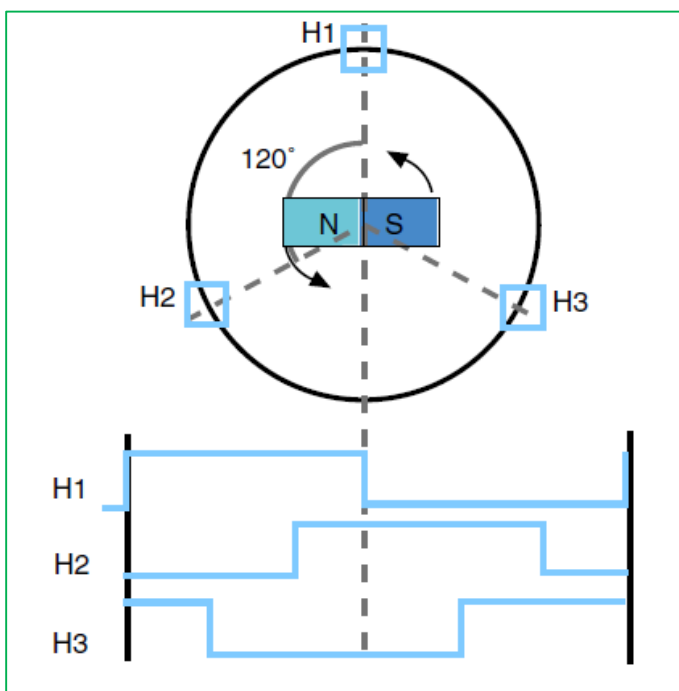


对于60度的Hall信号，可以任意调换三个信号中的任意一个即可得到和120度的处理相似，我们可以很方便使用软件处理。

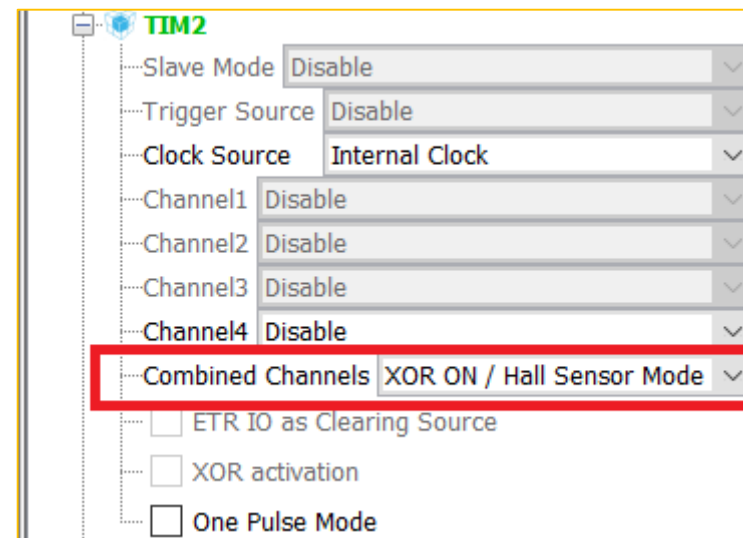
# STM32硬件Hall接口

29

- 在全系列STM32中都有硬件Hall接口(XOR输入)
- 可以每个Hall跳变沿都产生中断



CubeMx中的配置



# ST MC SDK5.x Hall传感器固件

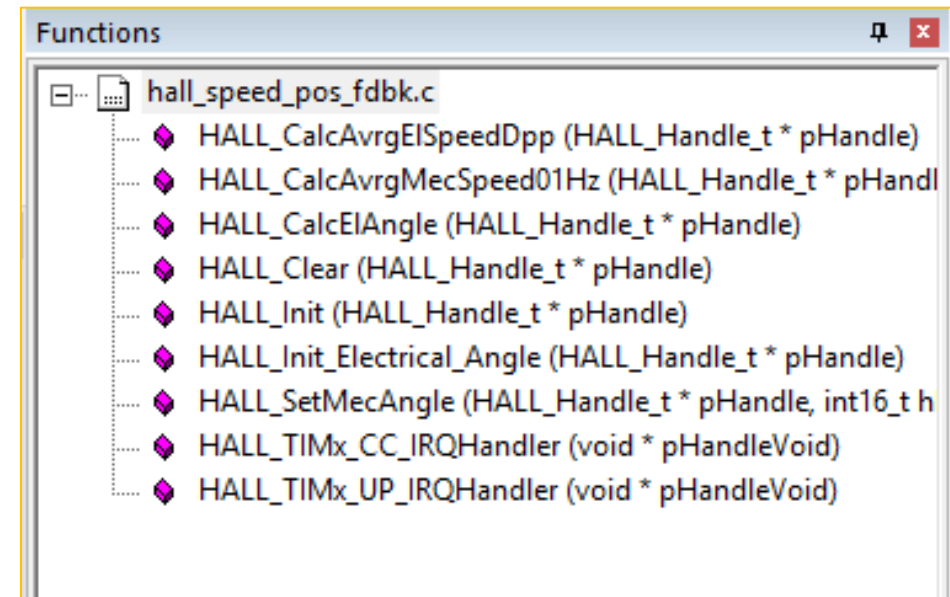
30

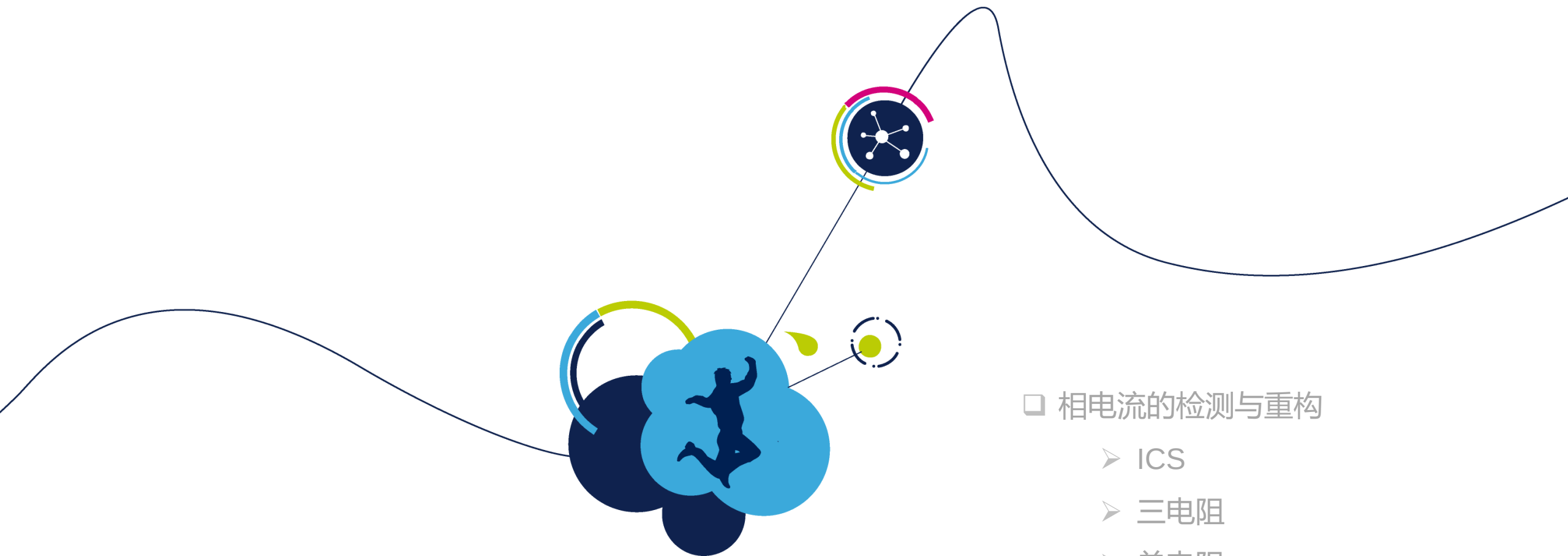
## ➤ 具体文件夹如下:

✓ xxx\MCSDK\_v5.2.0\MotorControl\MCSDK\MCLib\Any\Src

## ➤ 文件名称

✓ hall\_speed\_pos\_fdbk.c





# 转子位置、速度信息的获取

## □ 相电流的检测与重构

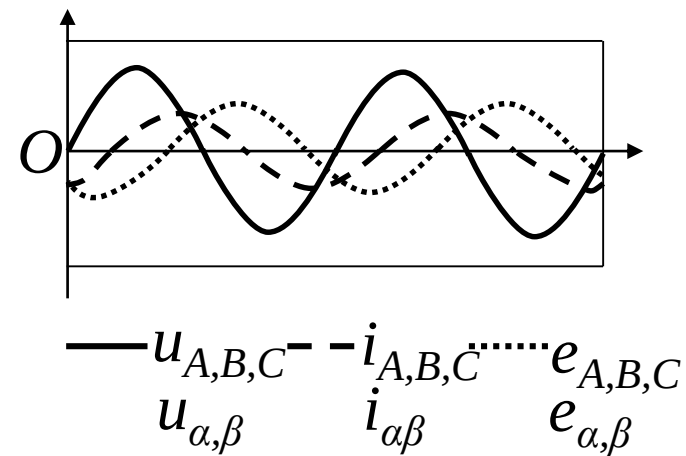
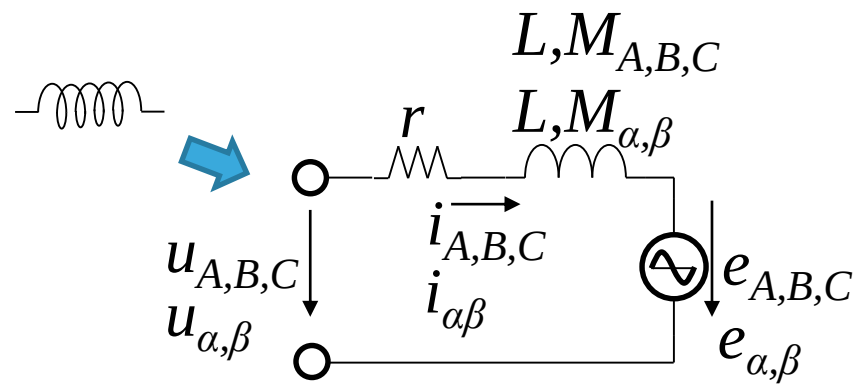
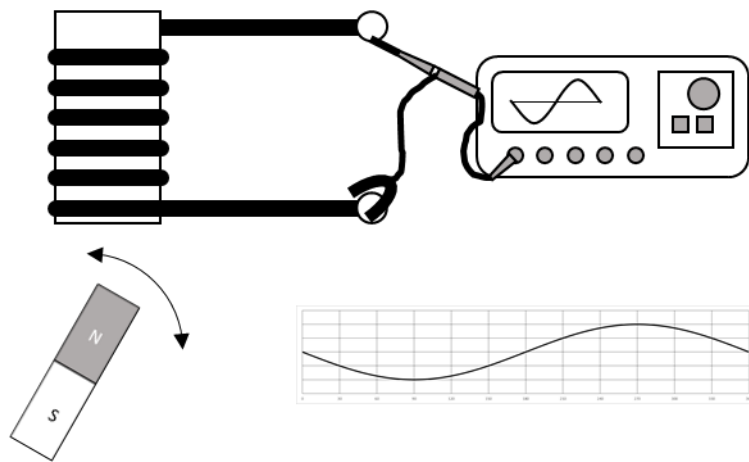
- ICS
- 三电阻
- 单电阻

## □ 转子位置、速度信息的获取

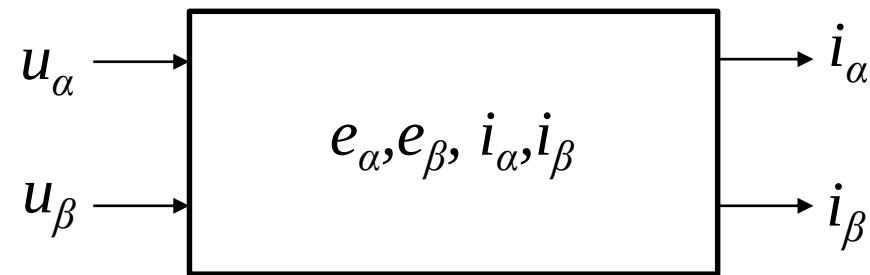
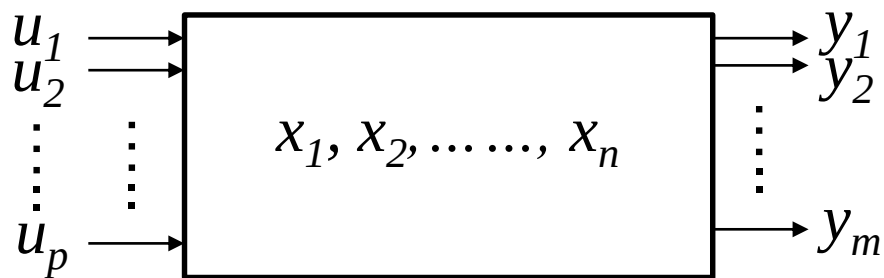
- 有位置传感器
- 无位置速度传感器

# 位置速度检测 — 观测器(1/13)

32

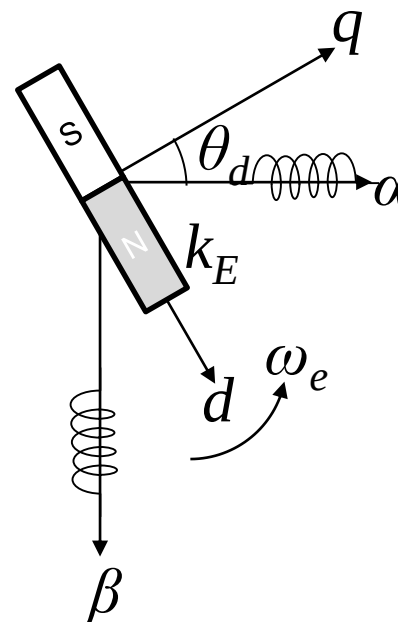
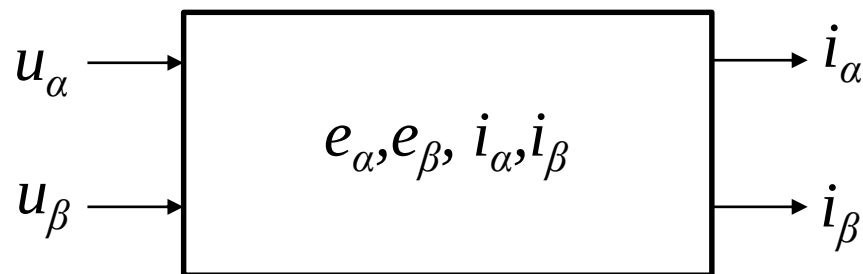
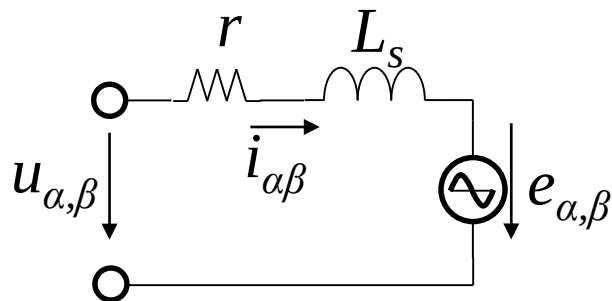


$$u \rightarrow i(\tau_e) \rightarrow \omega_r, \omega_e = p \cdot \omega_r \rightarrow e$$



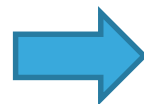
# 位置速度检测 — 观测器(2/13)

33



$$\begin{cases} e_{\alpha} = k_E \omega_e \cos(\omega_e t) \\ e_{\beta} = -k_E \omega_e \sin(\omega_e t) \end{cases}$$

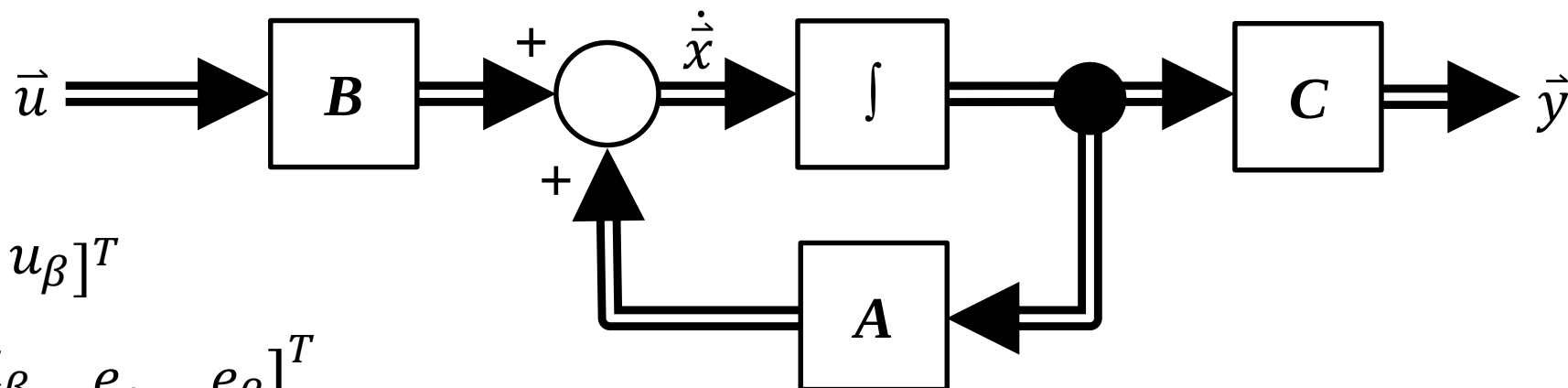
A linear model



$$\begin{cases} \frac{di_{\alpha}}{dt} = -\frac{r}{L_s} i_{\alpha} - \frac{e_{\alpha}}{L_s} + \frac{u_{\alpha}}{L_s} \\ \frac{di_{\beta}}{dt} = -\frac{r}{L_s} i_{\beta} - \frac{e_{\beta}}{L_s} + \frac{u_{\beta}}{L_s} \\ \frac{de_{\alpha}}{dt} = \omega_e e_{\beta} \\ \frac{de_{\beta}}{dt} = -\omega_e e_{\alpha} \end{cases}$$

# 位置速度检测 — 观测器(3/13)

34



$$\vec{u} = [u_\alpha \quad u_\beta]^T$$

$$\vec{x} = [i_\alpha \quad i_\beta \quad e_\alpha \quad e_\beta]^T$$

$$\dot{\vec{x}} = \begin{bmatrix} \frac{di_\alpha}{dt} & \frac{di_\beta}{dt} & \frac{de_\alpha}{dt} & \frac{de_\beta}{dt} \end{bmatrix}^T$$

$$\vec{y} = [i_\alpha \quad i_\beta]^T$$

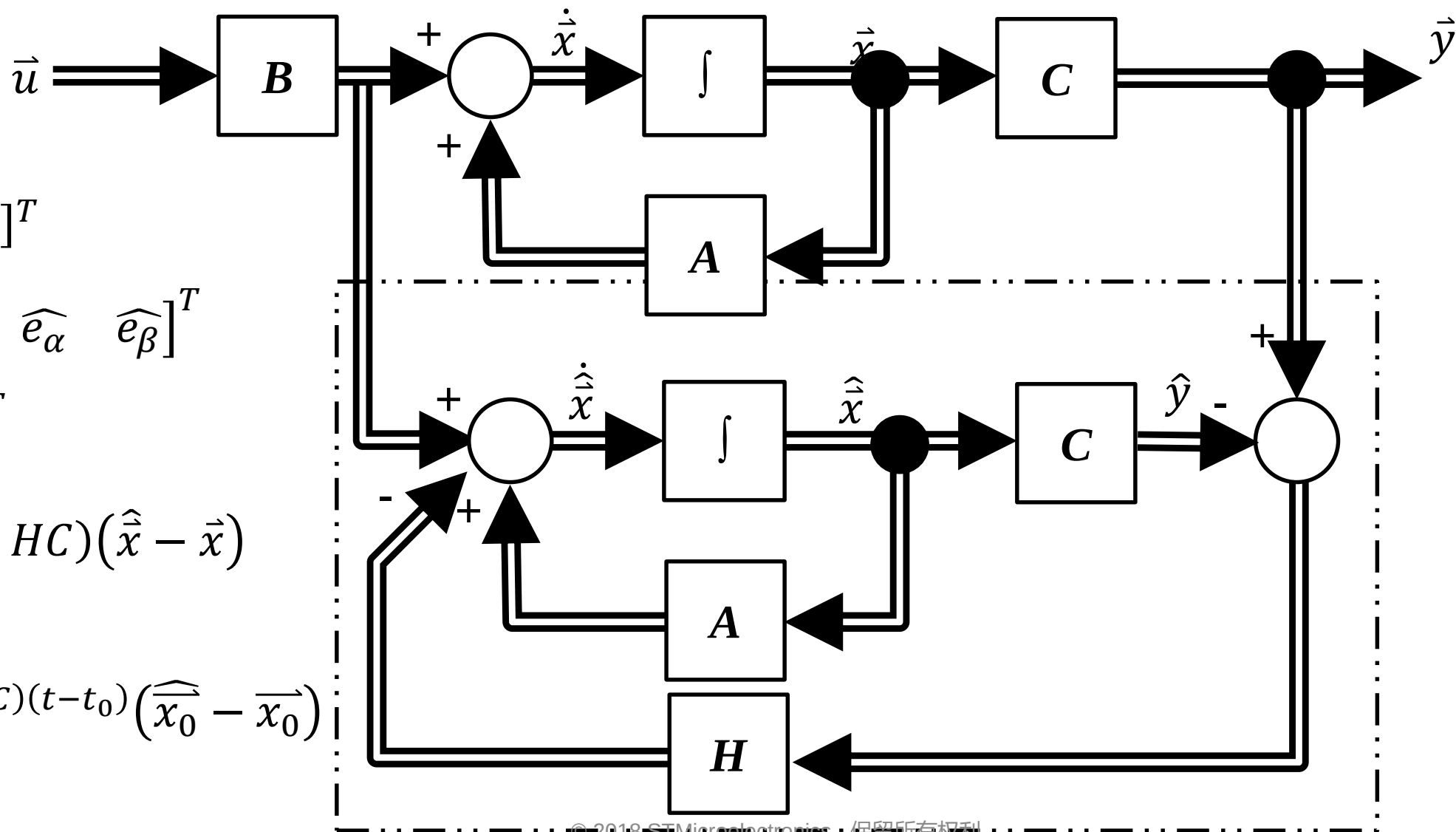
$$\begin{cases} \dot{\vec{x}} = A\vec{x} + B\vec{u} \\ \vec{y} = C\vec{x} \end{cases}$$

$$A = \begin{bmatrix} -\frac{r}{L_s} & 0 & -\frac{1}{L_s} & 0 \\ 0 & -\frac{r}{L_s} & 0 & -\frac{1}{L_s} \\ 0 & 0 & 0 & \omega_e \\ 0 & 0 & -\omega_e & 0 \end{bmatrix} \quad B = \begin{bmatrix} -\frac{1}{L_s} & 0 \\ 0 & -\frac{1}{L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

# 位置速度检测 — 观测器(4/13)

35



$$\vec{u} = [u_\alpha \quad u_\beta]^T$$

$$\hat{\vec{x}} = [\hat{i}_\alpha \quad \hat{i}_\beta \quad \hat{e}_\alpha \quad \hat{e}_\beta]^T$$

$$\hat{y} = [\hat{i}_\alpha \quad \hat{i}_\beta]^T$$

$$\dot{\hat{\vec{x}}} - \dot{\vec{x}} = (A + HC)(\hat{\vec{x}} - \vec{x})$$



$$\hat{\vec{x}} - \vec{x} = e^{(A+HC)(t-t_0)}(\hat{\vec{x}}_0 - \vec{x}_0)$$

# 位置速度检测 — 观测器(5/13)

36

$$\begin{cases} \dot{\vec{x}} = A\vec{x} + B\vec{u} \\ \vec{y} = C\vec{x} \end{cases} \xrightarrow{\text{离散化}} \begin{cases} \frac{\vec{x}[k] - \vec{x}[k-1]}{T_s} = A\vec{x}[k-1] + B\vec{u}[k-1] \\ \vec{y}[k] = C\vec{x}[k] \end{cases}$$

$$\begin{cases} i_\alpha[k] = \left(1 - \frac{rT_s}{L_s}\right) i_\alpha[k-1] - \frac{T_s}{L_s} e_\alpha[k-1] + \frac{T_s}{L_s} u_\alpha \\ i_\beta[k] = \left(1 - \frac{rT_s}{L_s}\right) i_\beta[k-1] - \frac{T_s}{L_s} e_\beta[k-1] + \frac{T_s}{L_s} u_\beta \\ e_\alpha[k] = e_\alpha[k-1] + \omega_e e_\beta[k-1] T_s \\ e_\beta[k] = e_\beta[k-1] - \omega_e e_\alpha[k-1] T_s \end{cases}$$

# 位置速度检测 — 观测器(6/13)

37

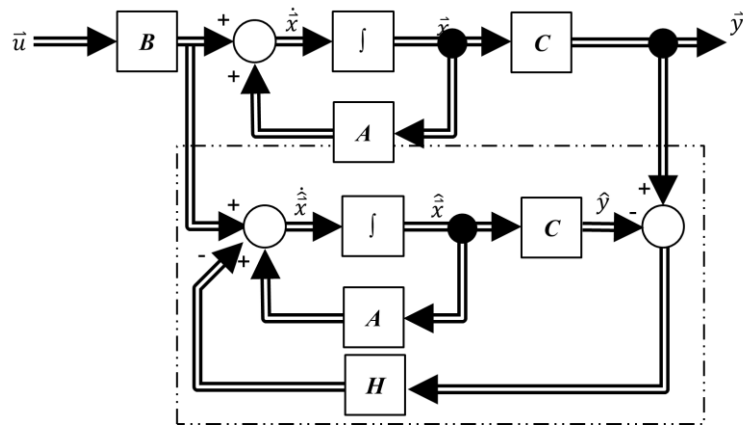
去耦(认为 $\omega_e=0$ ) 将简化马达模型

$$\begin{cases} i_\alpha[k] = \left(1 - \frac{rT_s}{L_s}\right) i_\alpha[k-1] - \frac{T_s}{L_s} e_\alpha[k-1] + \frac{T_s}{L_s} u_\alpha \\ e_\alpha[k] = e_\alpha[k-1] \end{cases}$$

$$A_{reduced} = \begin{bmatrix} 1 - \frac{rT_s}{L_s} & -\frac{T_s}{L_s} \\ 0 & 1 \end{bmatrix} \xrightarrow[\quad |\lambda I - A| = 0]{\text{特征值}} \begin{cases} \lambda_1 = 1 - \frac{rT_s}{L_s} \\ \lambda_2 = 1 \end{cases}$$

# 位置速度检测 — 观测器(7/13)

38



$$\dot{\hat{x}} = A\hat{x} + B\bar{u} + H(\hat{y} - \vec{y}) \quad H = \begin{bmatrix} h_1 & 0 \\ 0 & h_1 \\ h_2 & 0 \\ 0 & h_2 \end{bmatrix}$$

$$\begin{cases} \hat{i}_\alpha[k] = \left(1 - \frac{rT_s}{L_s} + h_1T_s\right) \hat{i}_\alpha[k-1] - \frac{T_s}{L_s} \hat{e}_\alpha[k-1] + \frac{T_s}{L_s} u_\alpha[k-1] + h_1T_s(-i_\alpha[k-1]) \\ \hat{i}_\beta[k] = \left(1 - \frac{rT_s}{L_s} + h_1T_s\right) \hat{i}_\beta[k-1] - \frac{T_s}{L_s} \hat{e}_\beta[k-1] + \frac{T_s}{L_s} u_\beta[k-1] + h_1T_s(-i_\beta[k-1]) \\ \hat{e}_\alpha[k] = \hat{e}_\alpha[k-1] + T_s\omega_e \hat{e}_\beta[k-1] + h_2T_s(\hat{i}_\alpha[k-1] - i_\alpha[k-1]) \\ \hat{e}_\beta[k] = \hat{e}_\beta[k-1] - T_s\omega_e \hat{e}_\alpha[k-1] + h_2T_s(\hat{i}_\beta[k-1] - i_\beta[k-1]) \end{cases}$$

# 位置速度检测 — 观测器(8/13)

39

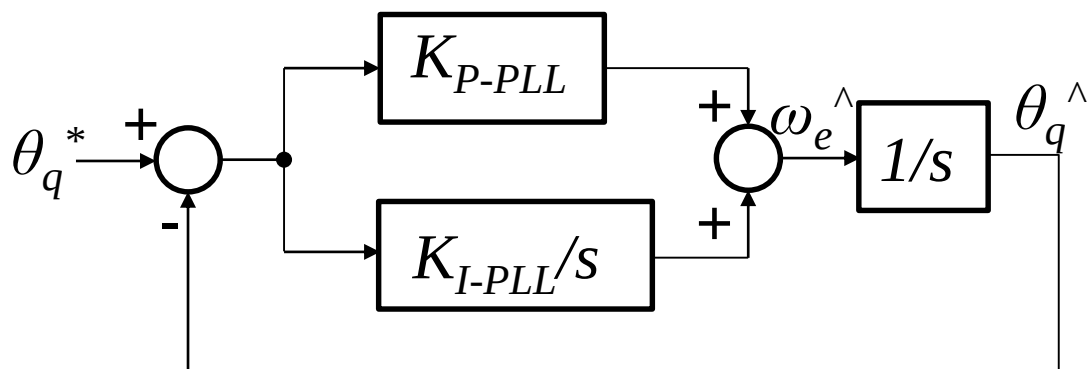
去耦(设定 $\omega_e=0$ ) 简化观测器模型

$$\begin{cases} \hat{i}_\alpha[k] = \left(1 - \frac{rT_s}{L_s} + h_1T_s\right) \hat{i}_\alpha[k-1] - \frac{T_s}{L_s} \hat{e}_\alpha[k-1] + \frac{T_s}{L_s} u_\alpha[k-1] + h_1T_s(-i_\alpha[k-1]) \\ \hat{e}_\alpha[k] = \hat{e}_\alpha[k-1] + h_2T_s(\hat{i}_\alpha[k-1] - i_\alpha[k-1]) \end{cases}$$

$$A_{reduced-obs} = \begin{bmatrix} 1 - \frac{rT_s}{L_s} + h_1T_s & -\frac{T_s}{L_s} \\ h_2T_s & 1 \end{bmatrix}$$
$$\begin{cases} \lambda_{1-obs} = \frac{\lambda_1}{k} \\ \lambda_{2-obs} = \frac{\lambda_2}{k} \end{cases} \quad k > 1$$
$$|\lambda I - A_{reduced-obs}| = (\lambda - \lambda_{1-obs})(\lambda - \lambda_{2-obs})$$
$$\begin{cases} h_1 = \frac{\lambda_{1-obs} + \lambda_{2-obs} - 2}{T_s} + \frac{r}{L_s} \\ h_2 = \frac{L_s(1 - \lambda_{1-obs} - \lambda_{2-obs} + \lambda_{1-obs}\lambda_{2-obs})}{T_s^2} \end{cases}$$

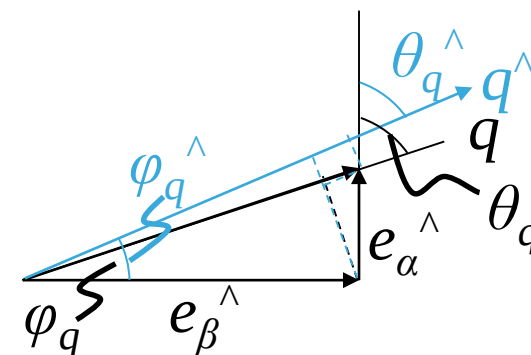
# 位置速度检测 — 观测器(9/13)

40



$$G(s) = \frac{Ts + 1}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\begin{cases} K_{P-PLL} = \omega_n \\ K_{I-PLL} = \frac{K_{P-PLL}\omega_n}{(2\zeta)^2} \end{cases}$$



$$e = k_E \omega_e = \sqrt{e_{\alpha}^{\wedge 2} + e_{\beta}^{\wedge 2}}$$

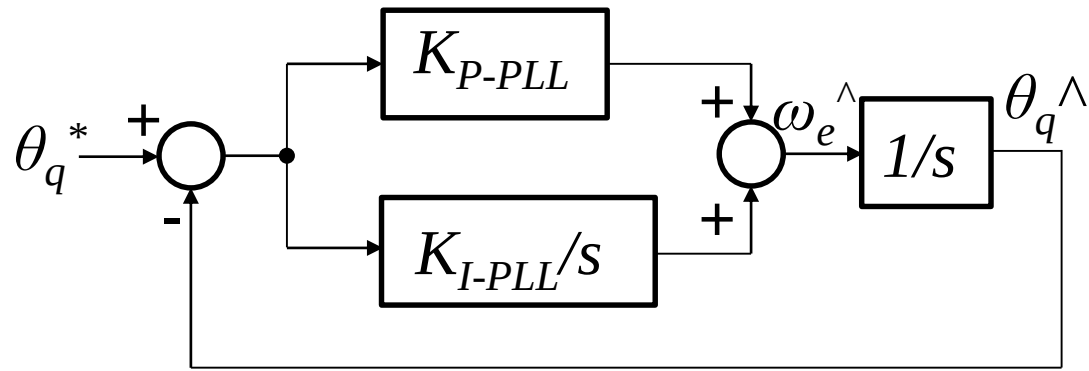
$$e \cdot \sin(\theta_q^{\wedge} - \theta_q) = e_{\beta} \cos(\theta_q^{\wedge}) - e_{\alpha} \sin(\theta_q^{\wedge})$$

↓ 如果  $\theta_q^{\wedge} - \theta_q$  比较小

$$\theta_q^{\wedge} - \theta_q \approx (e_{\beta} \cos(\theta_q^{\wedge}) - e_{\alpha} \sin(\theta_q^{\wedge})) / (k_E \omega_e)$$

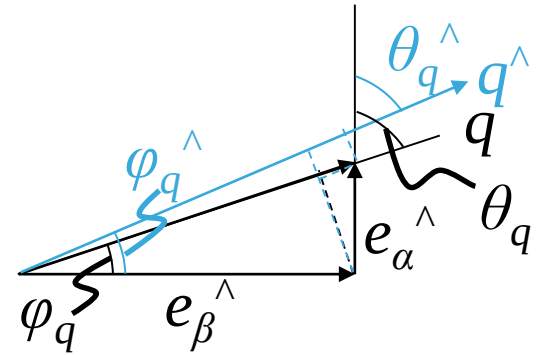
# 位置速度检测 — 观测器(10/13)

41

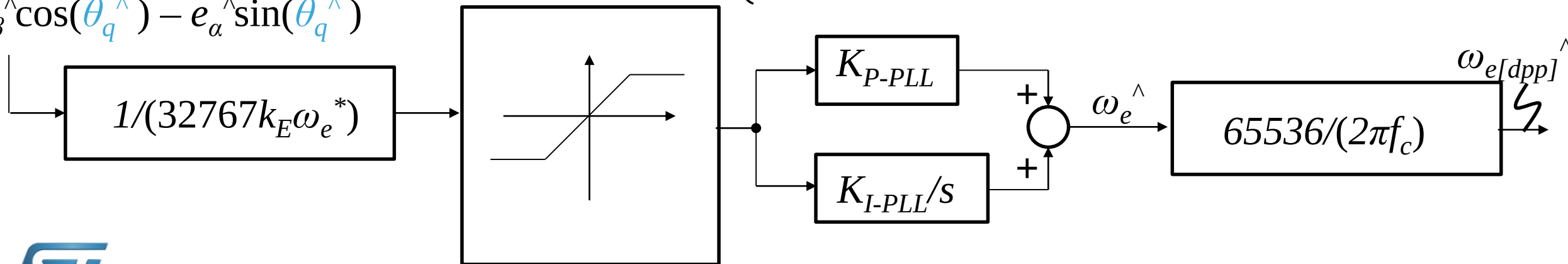


$$e = \sqrt{e_\alpha^2 + e_\beta^2} = k_E \omega_e^*$$

$$\begin{cases} K_{P-PLL} = \omega_n \\ K_{I-PLL} = \frac{K_{P-PLL} \omega_n}{(2\zeta)^2} \end{cases}$$

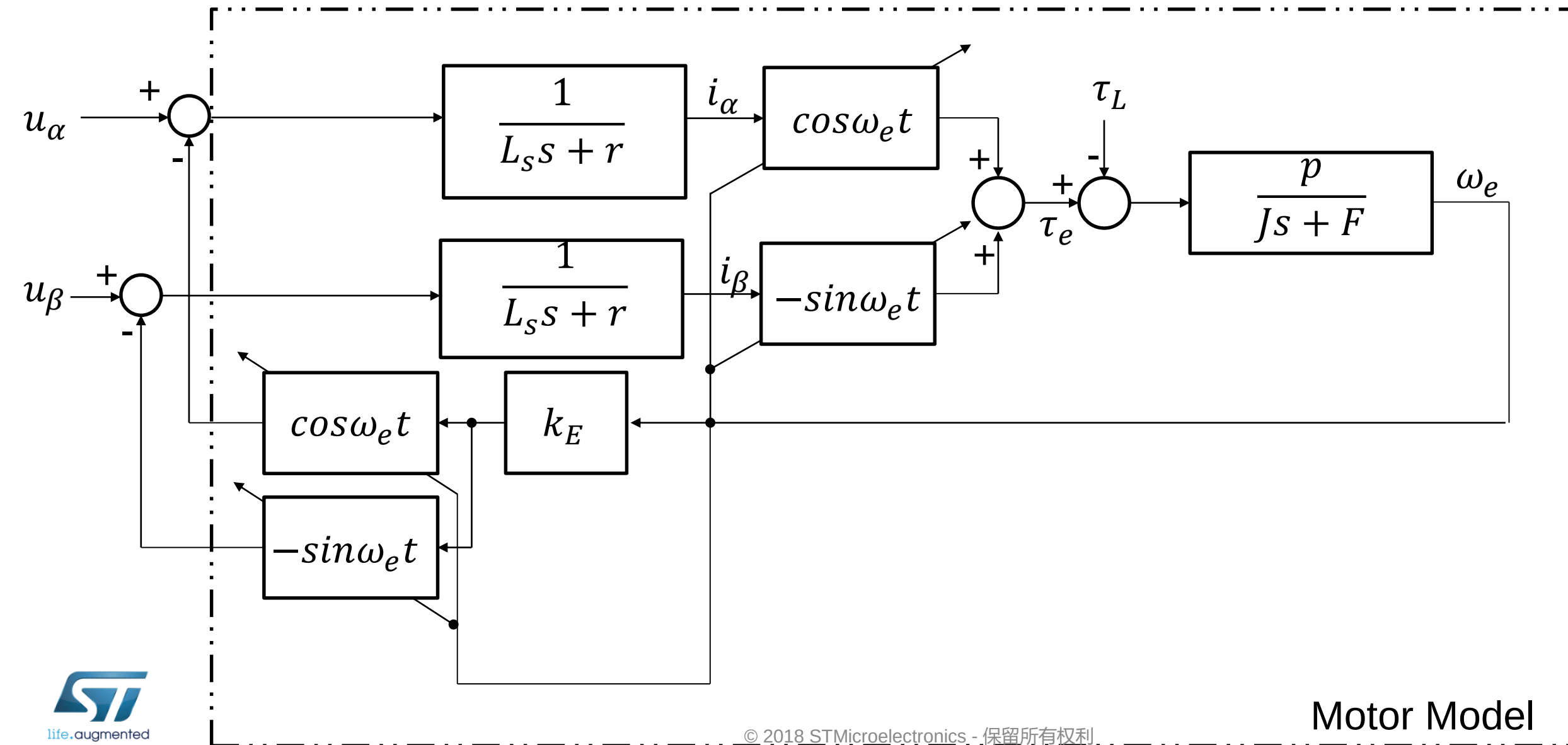


$$e_\beta^\wedge \cos(\theta_q^\wedge) - e_\alpha^\wedge \sin(\theta_q^\wedge)$$



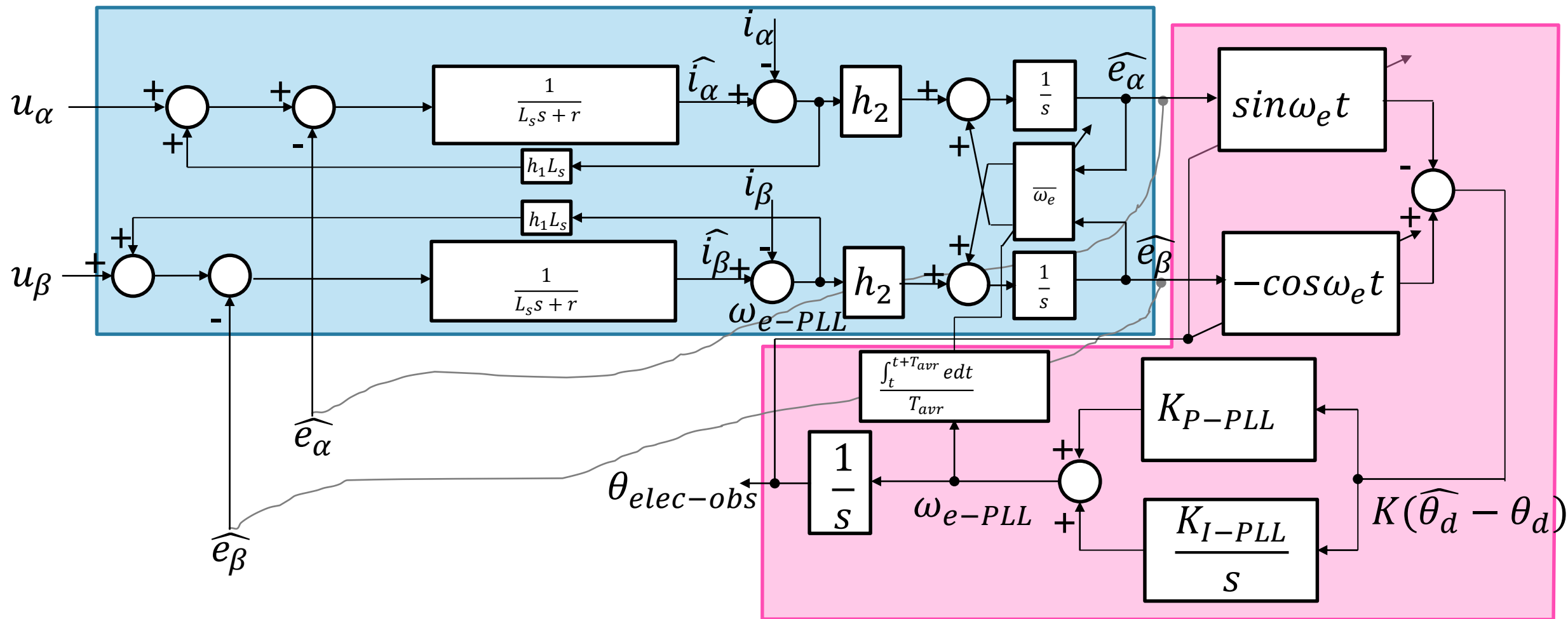
# 位置速度检测 — 观测器(11/13)

42



# 位置速度检测 — 观测器(12/13)

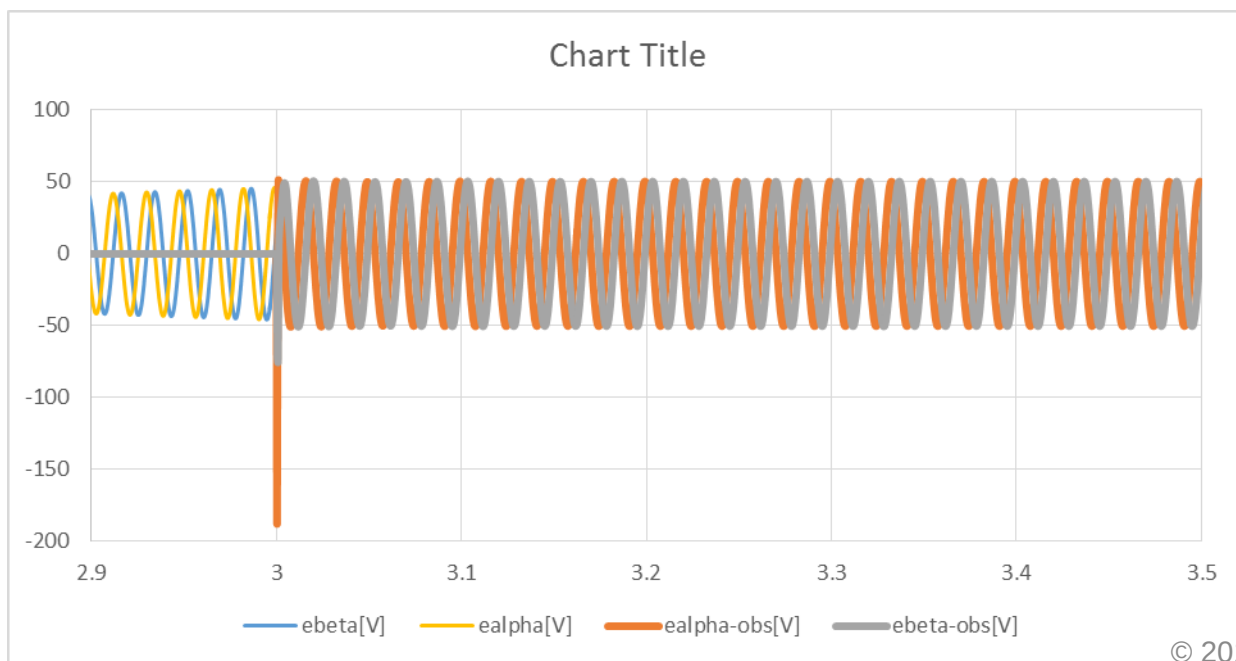
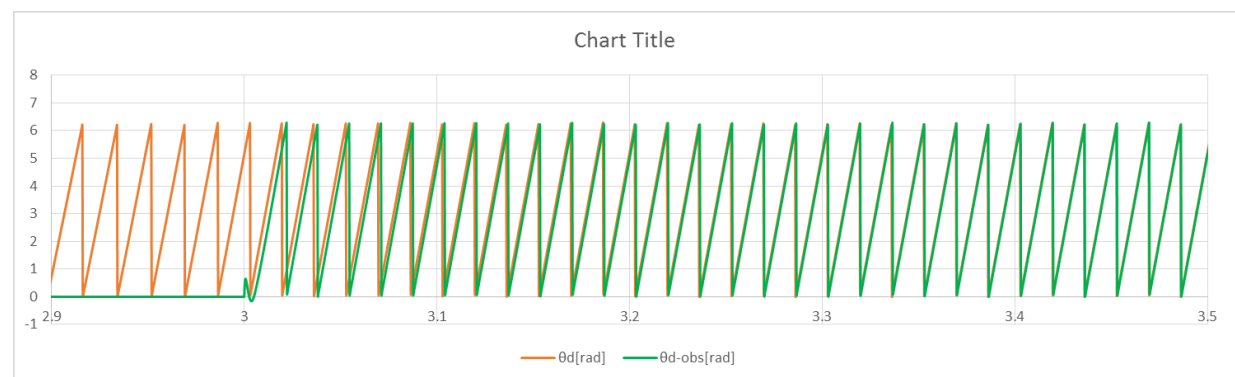
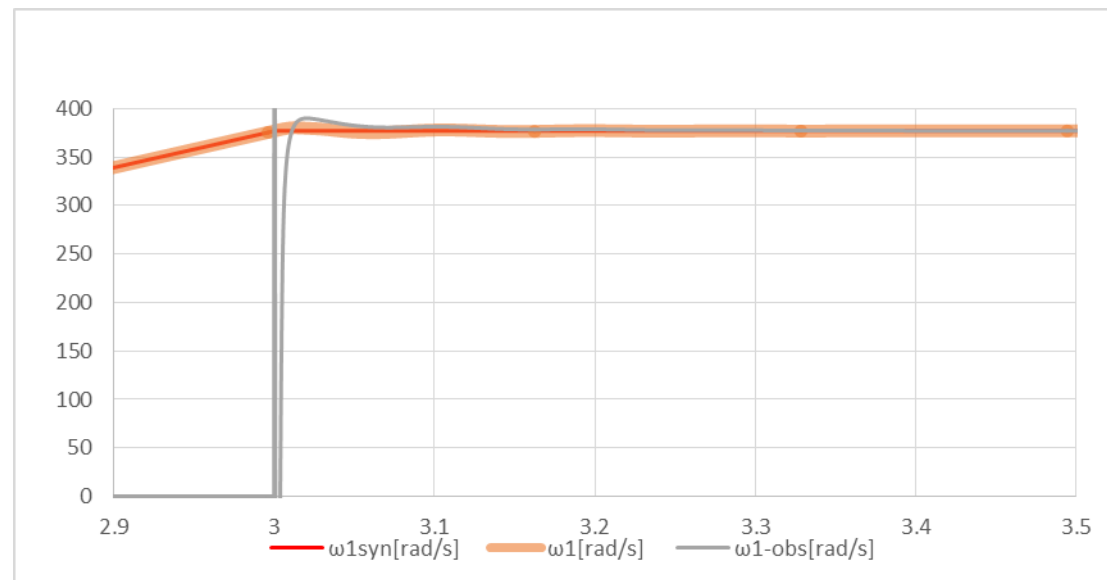
43



# 位置速度检测 — 观测器(13/13)

44

$$\begin{cases} \hat{i}_\alpha[k] = \left(1 - \frac{rT_s}{L_s} + h_1T_s\right) \hat{i}_\alpha[k-1] - \frac{T_s}{L_s} \hat{e}_\alpha[k-1] + \frac{T_s}{L_s} u_\alpha[k-1] + h_1T_s i_\alpha[k-1] \\ \hat{i}_\beta[k] = \left(1 - \frac{rT_s}{L_s} + h_1T_s\right) \hat{i}_\beta[k-1] - \frac{T_s}{L_s} \hat{e}_\beta[k-1] + \frac{T_s}{L_s} u_\beta[k-1] + h_1T_s i_\beta[k-1] \\ \hat{e}_\alpha[k] = \hat{e}_\alpha[k-1] + T_s \bar{\omega}_1 \hat{e}_\beta[k-1] + h_2T_s (\hat{i}_\alpha[k-1] - i_\alpha[k-1]) \\ \hat{e}_\beta[k] = \hat{e}_\beta[k-1] - T_s \bar{\omega}_1 \hat{e}_\alpha[k-1] + h_2T_s (\hat{i}_\beta[k-1] - i_\beta[k-1]) \end{cases}$$



# ST MC SDK5.x 观测器STO的固件

45

➤ 对于X-CUBE-MCSDK：观测器的固件以库的形式提供。

✓ xxx\MCSDK\_v5.2.0\MotorControlLib

➤ 头文件名如下：

✓ sto\_cordic\_speed\_pos\_fdbk.h

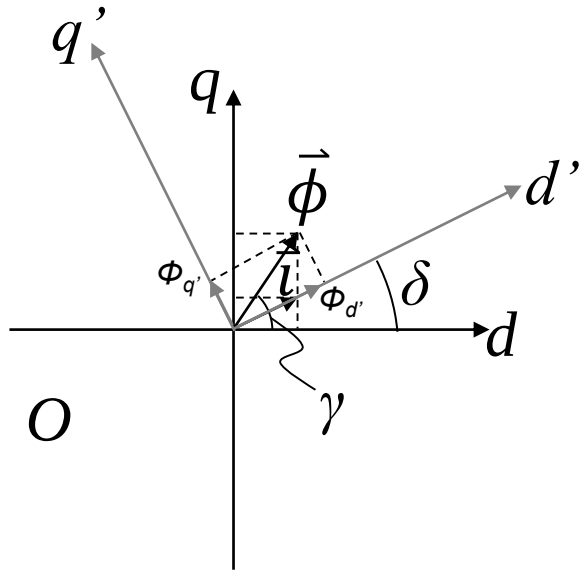
✓ sto\_pll\_speed\_pos\_fdbk.h

✓ sto\_speed\_pos\_fdbk.h

|                                                                                                                 |               |
|-----------------------------------------------------------------------------------------------------------------|---------------|
|  sto_cordic_speed_pos_fdbk.h | STO+Cordic头文件 |
|  sto_pll_speed_pos_fdbk.h    | STO+PLL头文件    |
|  sto_speed_pos_fdbk.h        | STO Handle定义  |

# 位置速度检测 — HFI(1/4)

46



$$\tan \delta = \frac{I_q}{I_d}, \tan \gamma = \frac{L_q I_q}{L_d I_d} = \frac{L_q}{L_d} \tan \delta$$

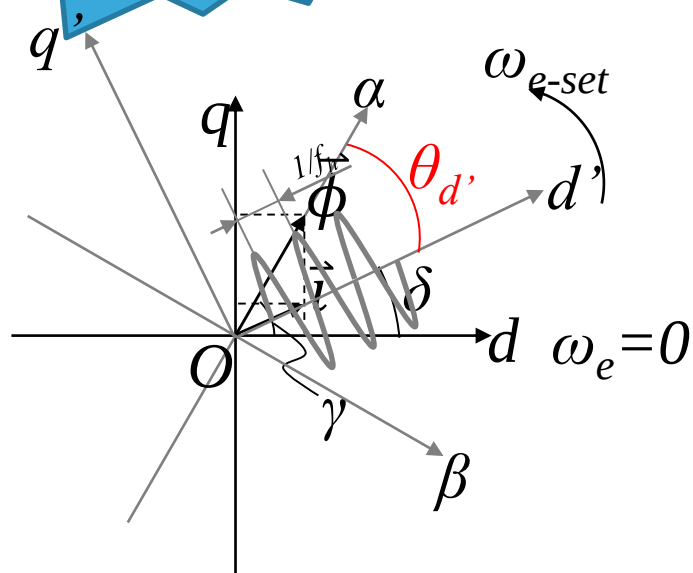
$$\tan(\gamma - \delta) = \frac{\tan \gamma - \tan \delta}{1 + \tan \gamma \tan \delta} = \frac{\left(\frac{L_q}{L_d} - 1\right) \tan \delta}{1 + \frac{L_q}{L_d} \tan^2 \delta} = \frac{(L_q - L_d) \sin 2\delta}{2[L_d + (L_q - L_d) \sin^2 \delta]}$$

$$|\vec{\phi}| = \sqrt{(L_d I_d)^2 + (L_q I_q)^2} = |\vec{i}| \sqrt{L_d^2 + (L_q^2 - L_d^2) \sin^2 \delta}$$

ST 专利  
US932563 B1

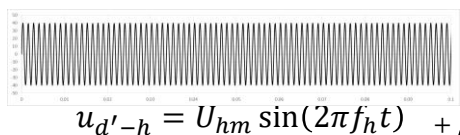
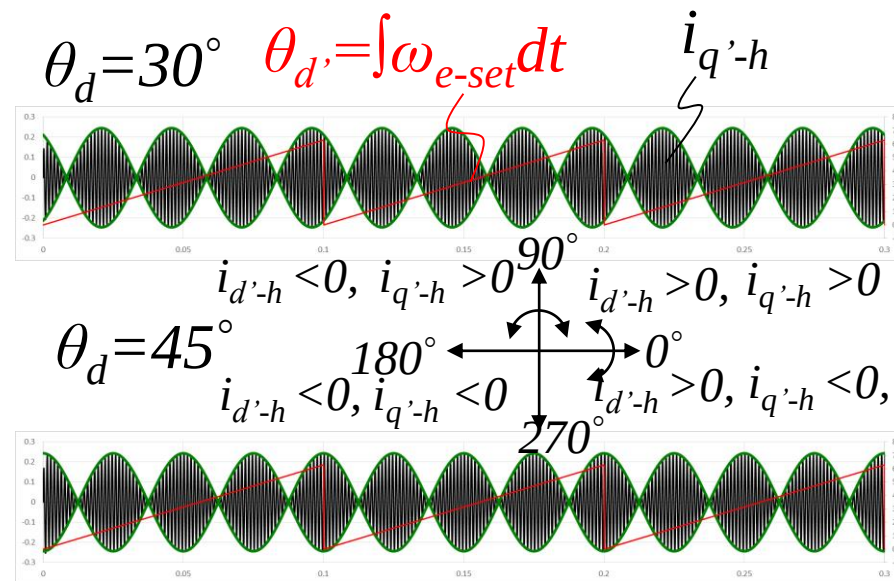
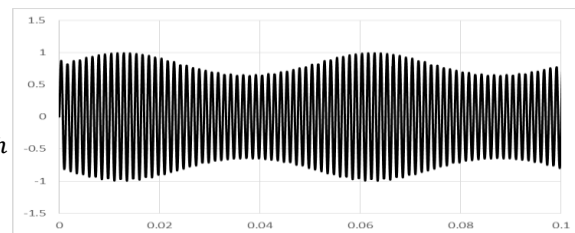
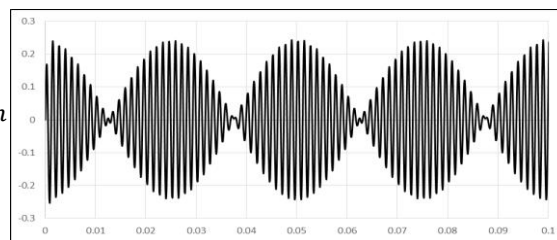
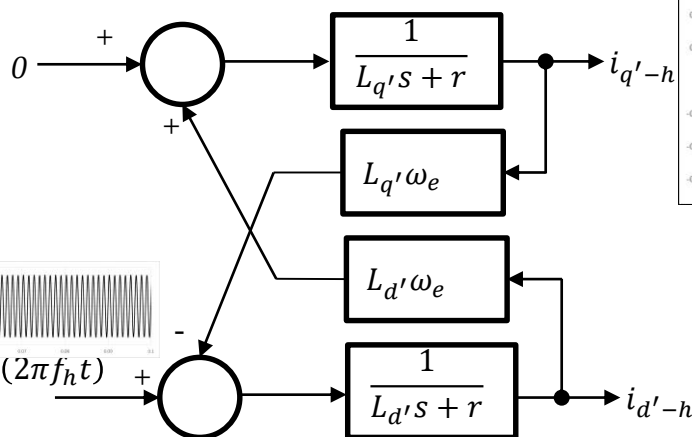
# 位置速度检测 — HFI(2/4)

47



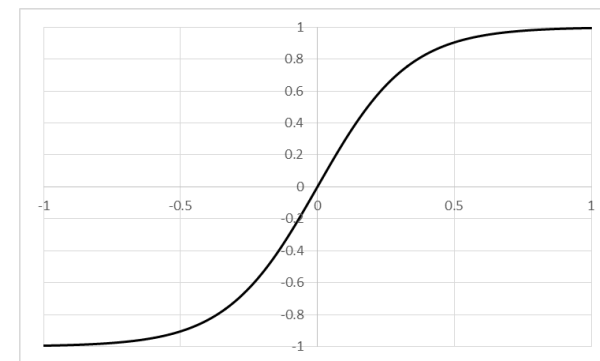
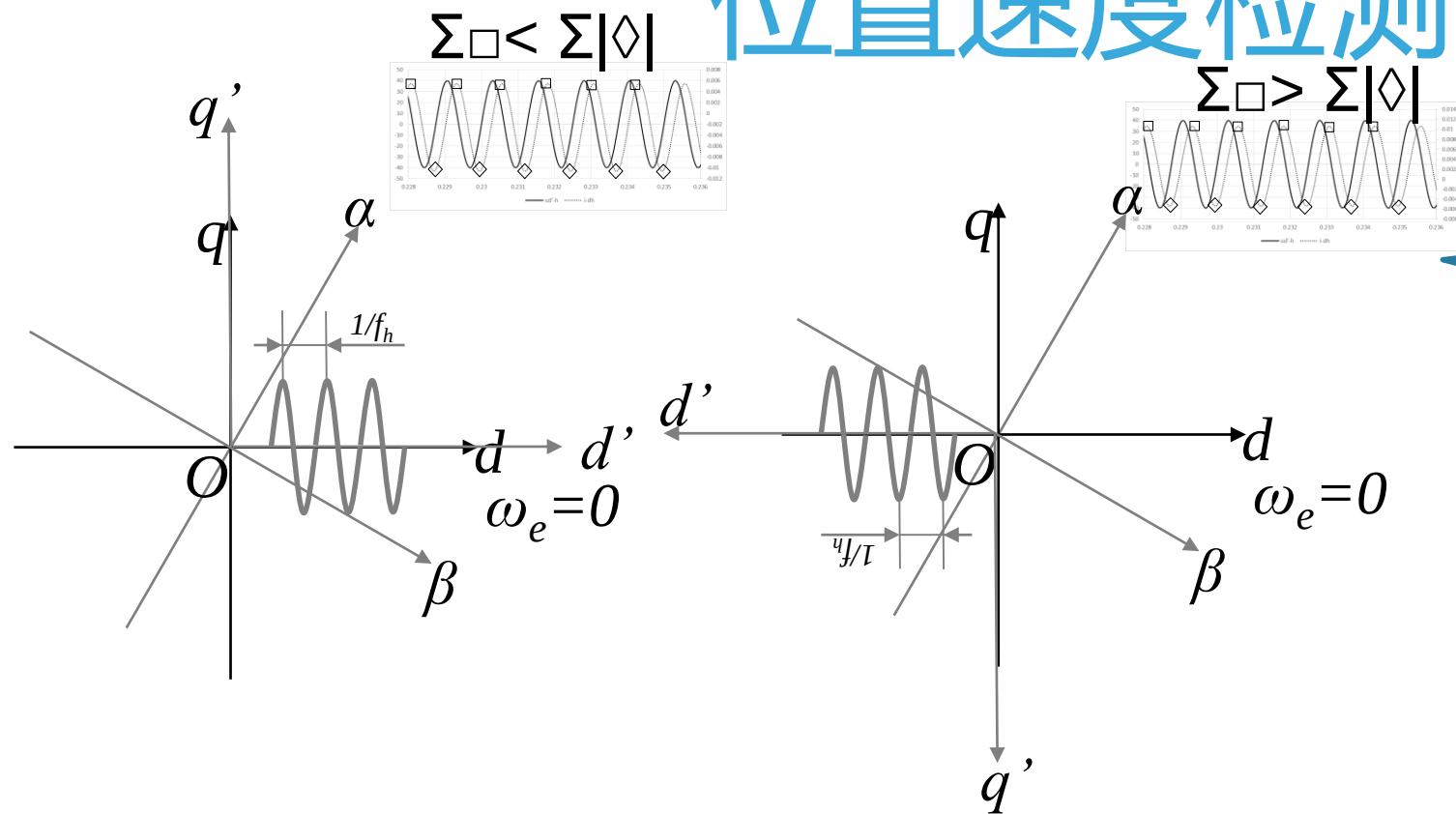
$$|\vec{\phi}| = \sqrt{(L_d I_d)^2 + (L_q I_q)^2} = |\vec{i}| \sqrt{L_d^2 + (L_q^2 - L_d^2) \sin^2 \delta}$$

$$|\vec{\phi}| = I_m \sin(2\pi f_h t) \sqrt{L_d^2 + (L_q^2 - L_d^2) \sin^2(\omega_e t + \delta_0)}$$

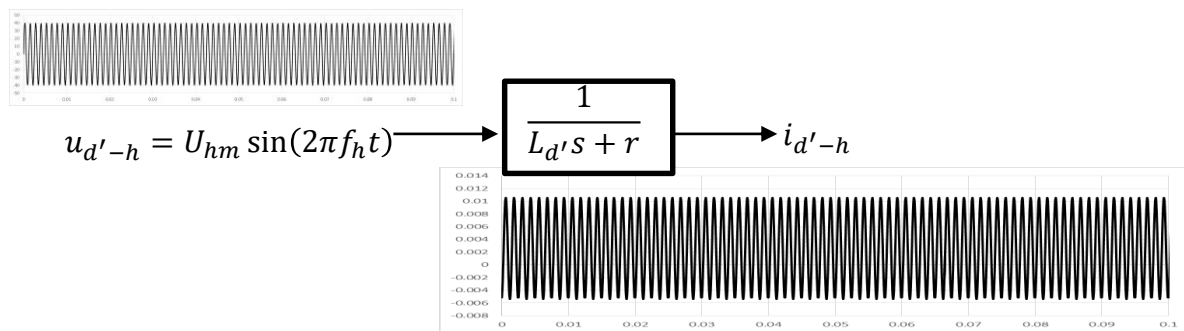


# 位置速度检测 — HFI(3/4)

ST 专利  
US932563 B1

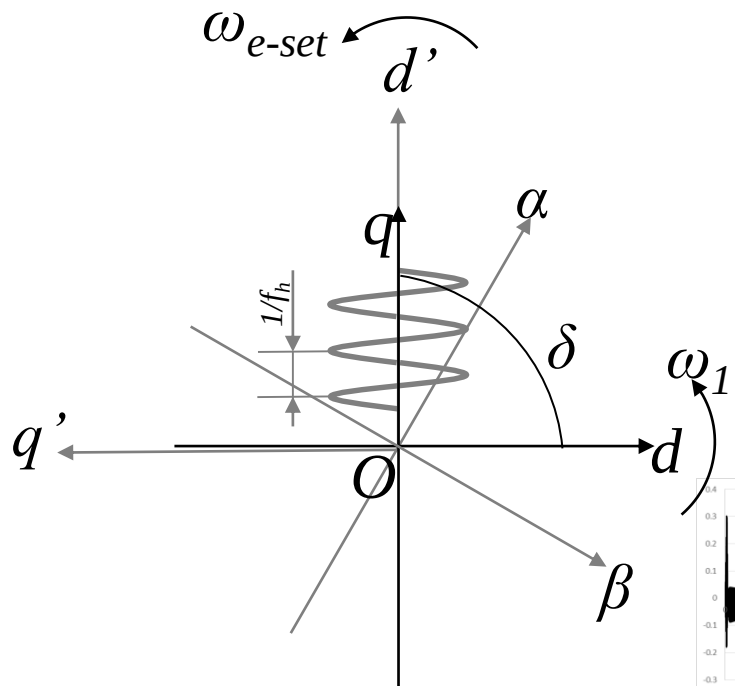


$\Phi$ - $i$  曲线



# 位置速度检测 — HFI(4/4)

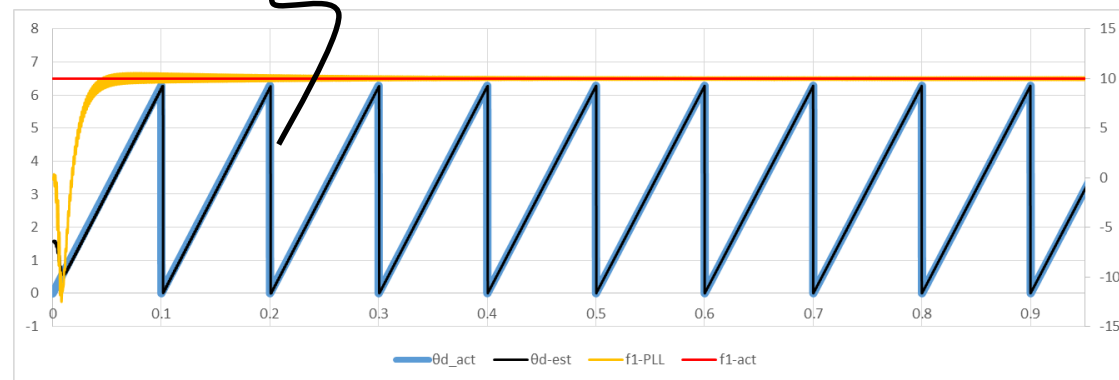
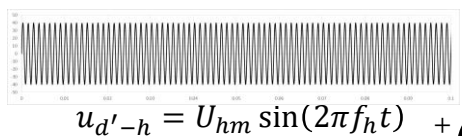
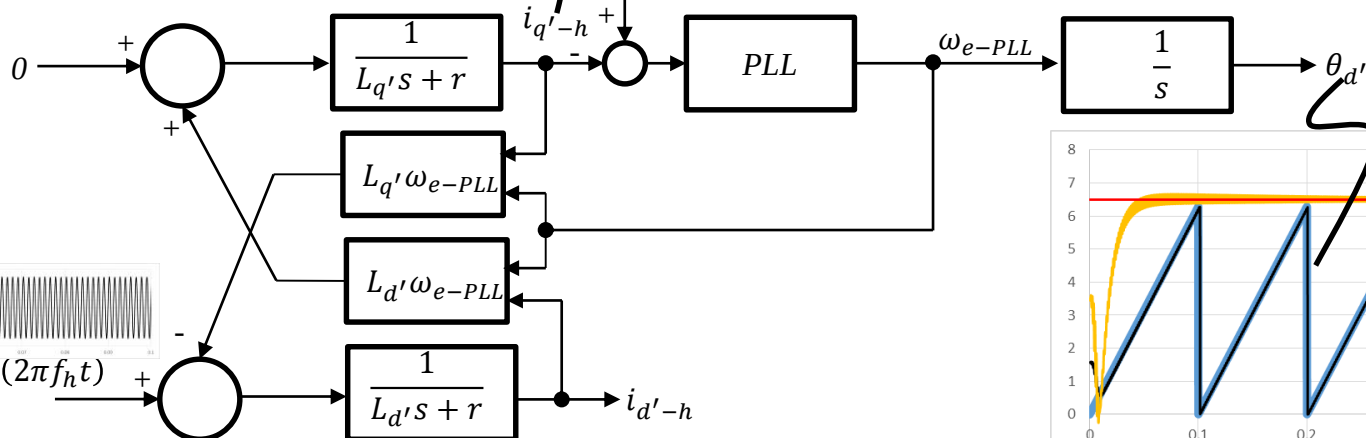
49



当电机开始转动后，保持 $\delta=0$ ， $d'$ 轴与 $q$ 轴重合，那么 $i_{q'-h}=0$ ，  
也就是如果保持 $i_{q'-h}=0$ ， $d'$ - $q'$  旋转坐标系的速度等于转子速度 $\omega_e = \omega_{e-set}$ 。



使用PLL,使得 $i_{q'-h}=0$



# Releasing your creativity

50



- Thank you -

## 重要通知 – 请仔细阅读

意法半导体公司及其子公司（“ST”）保留随时对ST 产品和/ 或本文档进行变更、更正、增强、修改和改进的权利，恕不另行通知。买方在订货之前应获取关于ST 产品的最新信息。ST 产品的销售依照订单确认时的相关ST 销售条款。

买方自行负责对ST 产品的选择和使用， ST 概不承担与应用协助或买方产品设计相关的任何责任。

ST 不对任何知识产权进行任何明示或默示的授权或许可。

转售的ST 产品如有不同于此处提供的信息的规定，将导致ST 针对该产品授予的任何保证失效。

ST 和ST 徽标是ST 的商标。所有其他产品或服务名称均为其各自所有者的财产。

本文档中的信息取代本文档所有早期版本中提供的信息。

## 版权声明

本文档为意法半导体公司及其子公司（“ST”）版权所有，未经ST允许不得复制、修改、转发或应用于商业目的。